

Subdued Crude Paving the Way for Less-Lewd Renewed Car Food

Senior Design II Report 2
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1. Project Charter

Title	Goal	In Scope	Out of Scope
Subdued Crude Paving the Way for Less-Lewd Renewed Car Food	To design a full PFD, mass balance, and equipment design for a biodiesel production plant	Development of a complete PFD, utility usage, feasibility of design	• Process ease • Location impact • Capital investment • Safety concerns • Cost of equipment and utilities
Deliverables		Team Members	
• Design Plan • Complete detailed PFD • Detailed mass balance • Updated BFD • Complete equipment list • Preliminary utility usage		Jake Proctor Milos Markovic Kellen Krupa Cheryl Slykas	

2. Design Plan

Report 2					
Task ID	Task	Start Date	Duration (Days)	Completion Date	Assignee
Lostics					
1	Weekly Meetings	1-Feb	29	29-Feb	All Members
4	Project Charter and Design Plan	1-Feb	1	2-Feb	Jake
5	Equipment sizing reseach	1-Feb	2	3-Feb	Jake and Milos
6	Mass balance preparation	1-Feb	2	3-Feb	Cheryl and Kellen
7	Aspen simulation preparation	1-Feb	4	5-Feb	All Members
8	Layout / Format	1-Feb	1	2-Feb	All Members
Execution					
Updates From Last Report					
9	Update BFD	4-Feb	2	6-Feb	Kellen and Cheryl
10	Update PFD in detail	5-Feb	5	10-Feb	Jake
11	Update market analysis	4-Feb	2	6-Feb	Cheryl
12	Update introduction	4-Feb	2	6-Feb	Kellen
13	Update process description	4-Feb	2	6-Feb	Milos
New Items					
14	Detailed mass balance section 1	20-Feb	4	24-Feb	Cheryl and Kellen
15	Detailed mass balance section 2	20-Feb	4	24-Feb	Kellen and Cheryl
16	Detailed mass balance section 3	20-Feb	4	24-Feb	Cheryl and Kellen
17	ASPEN simulation	6-Feb	14	20-Feb	Milos and Team
18	Equipment sizing section 1	21-Feb	4	25-Feb	Jake and Milos
19	Equipment sizing section 2	21-Feb	4	25-Feb	Milos and Jake
20	Equipment sizing section 3	21-Feb	4	25-Feb	Jake and Milos
21	Preliminary utility summary	24-Feb	3	27-Feb	Kellen and Cheryl
22	Hand calcuatlions and comparison	23-Feb	2	25-Feb	Kellen and Cheryl
Wrap-Up					
23	Section Unification	27-Feb	1	18-Feb	All Members
24	Proofreading / Plagiarism Check	28-Feb	1	29-Feb	All Members
25	Reference Page	26-Feb	3	29-Feb	All Members

3. Introduction

3.1. Background and Objective

In this report, the goal is to propose the most optimal process for manufacturing quality biodiesel from used cooking oil. The product can be used as B100, 100% biodiesel, or blended with petrol-diesel. Basing operations in the midwestern United States in the state of Illinois provides a domestic supply of used cooking oil and a large market for biodiesel. The United States is the global leader in biodiesel production, producing 6.9 billion liters in 2018. [1] Hundreds of environmental service companies that collect and filter waste oil are located in the United States, with several located in Illinois. Mahoney Environmental, Filta Environmental Kitchen Solutions, and Green Grease Environmental are a few examples of Illinois companies that sell refined waste cooking oil nationwide. [2]

Energy demands coincide with the rise in global population, which is expected to reach 11 billion people by the end of the twenty-first century. [3] Fossil fuels are a primary source of energy for the world accounting for 80% of the total energy consumption. [4] In the United States, the transportation sector accounted for 29% of greenhouse gas emissions from 1990 to 2017. Demand for fossil fuels will increase 124%, despite current projections that crude oil and gas will be depleted in fifty years. [5,6] Another major concern is the effects of fossil fuels on global warming. The burning of fossil fuels increases the level of greenhouse gases, such as carbon dioxide and nitrogen oxides, in the atmosphere, where the gases trap heat and cause the planet to warm. This rise in temperature contributes to more extreme weather, such as heat and cold waves and higher category hurricanes. [6] Also, the presence of these gases in the air and water causes acid rain and the acidification of the ocean, all of which are responsible for negative changes in ecosystems. [6, 7]

Modern fuel usage causes energy, environmental, and economic issues that have led to research into alternative energies that are sustainable and cost effective. Biodiesel, or fatty acid methyl esters (FAME), is one renewable fuel proven to provide energy security and economic stability and lessen environmental impacts as shown in Table 1. It can be used as a replacement or additive to petroleum diesel. [8] It reduces dependence on foreign fuel sources because it can be produced domestically from a variety of feedstocks, including vegetable oil, animal fats, and recycled restaurant grease. [9] There are government tax incentives that encourage companies to produce biodiesel. This is important for local economies as it creates jobs and stability in the area. [10] Additionally, biodiesel is nontoxic, biodegradable, and a cleaner burning fuel. [8] Since it contains very low sulfur content, it reduces fuel contributions to acid rain. Also, burning biodiesel means less particulate matter pollution and less smog. [11] By using biodiesel, hydrocarbon emissions are reduced by 67% and carbon emissions by 78.5%. [10, 11]

Table 1. Comparison of Properties of Different Fuel Fluids [12]

Fuel Property	Gasoline	No.2 Diesel	Iso-Octane	Methanol Biodiesel	Ethanol Biodiesel
Cetane Number	-	50	-	5	8
Octane Number	96	-	100	112	107
Auto-Ignition Temp (F)	700	600	494	867	631
Latent Heat of Vaporization (BTU/lb)	150	95	112	507	391

As shown in table 1 above, the physical properties of different types of fuels is shown. When comparing the different physical properties. The cetane number tells us how long of a combustion speed that type of fuel has when combusting. The methanol biodiesel has a much smaller cetane number making the combustion faster with less time needed for combustion. The other benefit of biofuels over regular diesel or gasoline is that the measure of engine performance before detonation (aka. Octane number) is much higher meaning more pressure can be put on the engine before it explodes. Biodiesel also comes with higher auto-ignition temperatures meaning that a higher temperature is required before the fuel begins to ignite itself. All these characteristics support fuel source that is biodiesel. Coupled with the higher capacity to vaporize, biodiesel also proves itself to be higher in energy worth for the combustion process.

One abundant feedstock used in the production of biodiesel is used cooking oil. In 2018, 1.4 billion pounds were used, with eight pounds of cooking oil for every one gallon of biodiesel. [13] Reusing a waste product such as used cooking oil to create biodiesel can reduce health and environmental issues associated with waste removal and storage. Fats, oils, and grease (FOG) solidify in drains and sewer lines, causing restrictions to waste flow. This accumulation of FOG, along with the presence of roots and debris in the sewer system, reduces the capacity in the sewer lines. As a result, a substantial accumulation causes sewage spills in the streets or backups into people's homes and businesses. According to the U.S. Environmental Protection Agency (EPA), over 40,000 sanitary sewer overflows occur each year and 47% are related to FOG blockages. [14] To avoid this problem, disposal of FOG into trash is highly encouraged, but even this creates more issues. Once transported to a landfill, the triglycerides that comprise FOGs naturally decompose to carbon dioxide and methane and enter the atmosphere, contributing to global warming. [15] In 2017, 14.1% of human related methane emissions were from municipal solid waste landfills. [16]

3.2. Market Analysis

To determine whether producing biodiesel from used cooking oil will be successful in Illinois, United States, it is essential to analyze the global and local markets including evaluating the industries utilizing biodiesel and whether the supply already meets the demand. Feedstock pricing and availability, product pricing, government incentives, competitors, and market drivers will also be examined to estimate the success of this project.

Global Market Overview:

Biodiesel is a clean-burning, renewable source of fuel that takes triglycerides and methanol and produces fatty acid methyl esters (biodiesel) and glycerin and is used in many industries including the automotive/transportation, marine, railway, and power industries. It can also be blended with traditional petrol fuels to make biodiesel blends. See Figure 1.

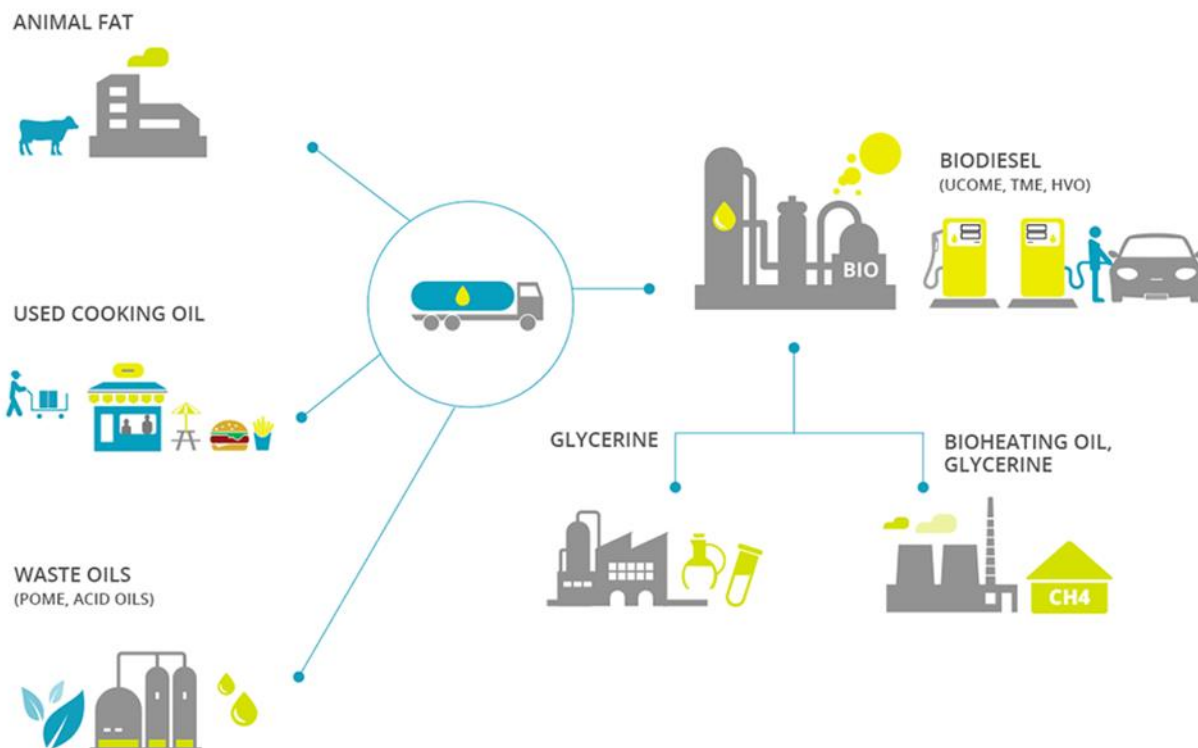


Figure 1: Value chain for production of biodiesel [17]

As fossil fuel reserves continue to deplete, the use of biodiesel is helping various countries decrease their dependency on foreign oil reserves because it is produced locally. Many governments provide incentives for the production and use of biodiesel. Biodiesel also reduces greenhouse gas emissions and is biodegradable, thus also complying to strict government environmental regulations. All of these are factors driving the biodiesel markets. [18]

The Transparency Market Research has determined that the global market for biodiesel is likely to grow with an expected 6.9% compound annual growth rate (CAGR) between the years 2017 and 2026. By 2026, the global market for biodiesel is projected to bring about \$64,497.40 million U.S. dollars in revenue. [18] The countries contributing the largest to biodiesel production in 2018 include the European Union, mainly Germany and France, the United States, Brazil and Argentina. See Figure 2. In 2018, 41.2 million tonnes of biodiesel was produced globally. [19]

Biodiesel production in key countries in 2018

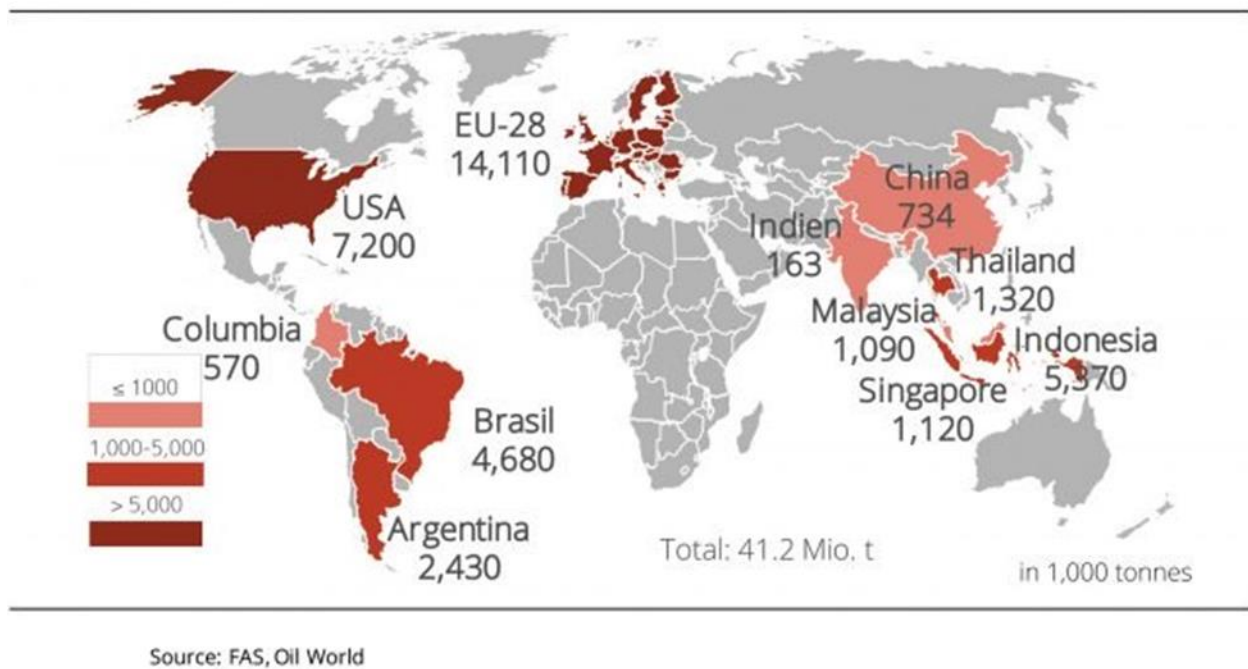


Figure 2. Countries contributing to the production of biodiesel in 2018 with corresponding production amounts in 1,000 tonnes [19]

“In 2001, total world biodiesel consumption was about 0.3 billion gallons. In 2016, about 9.3 billion gallons of biodiesel were consumed in at least 56 countries, and 58% was consumed in five countries.” [20] See Table 2. The United States ranked number 2 in total consumption of biodiesel.

Table 2. World biodiesel consumption in 2016 by country [20]

	Billion gallons	Share of world total
World total	9.3	
United States	2.1	22%
Brazil	1.0	10%
France	0.9	10%
Indonesia	0.8	9%
Germany	0.7	7%
All others	3.9	

US Market Overview.

Regulatory Drivers:

The United States biodiesel industry's growth has been steady and sustainable over the past 15 years, with support from the biodiesel tax incentive and the RFS (Renewable Fuel Standard). "Regulations passed by congress since 2007 have driven the commercial availability and sustainability of biodiesel." [21] The Energy Independence and Security Act of 2007 has been extended through 2022 and requires the use of renewable biofuels in the United States diesel and gasoline supply including biodiesel. The EPA set renewable volume obligations (RVOs) which obligates parties to blend sufficient volumes of renewable fuels to meet annual obligations. [21]

The Biodiesel Tax Credit was enacted in 2005 with the purpose of making United States fuel blenders sell biodiesel. It provided a credit of \$1 dollar per gallon against their U.S. federal tax liability. Even though this tax credit has expired multiple times since 2005, it has always been extended except in 2018. Without the Biodiesel Tax Credit, the industry could not have covered production costs from 2016 through the first quarter of 2019, and thus operated below industry breakeven point. The Biodiesel Tax Credit over the 2015-2018 period has been able to get producers above the breakeven point. With the absence of the credit in 2018-2019, biodiesel producers faced an increased risk. President Donald Trump extended this tax credit in December of 2019 for 5 years to keep biodiesel competitive in the fuel market and support expansion of the market. [21]

Supply and Demand:

The domestic US demand for biodiesel increased from about 0.26 million gallons to 1.8 billion gallons. This is a CAGR of 27% from 2010-2018. The United States exports fluctuated about 100,000 gallons, which was a decline of 1% CAGR from 2010 to 2018. The overall U.S. demand for biodiesel increased from 368 million gallons to 1.9 billion gallons, a 23% increase in CAGR during the 2010-2018 period. [22] See Figure 3. The rising demand in the United States makes it a good target for the proposed plant location.

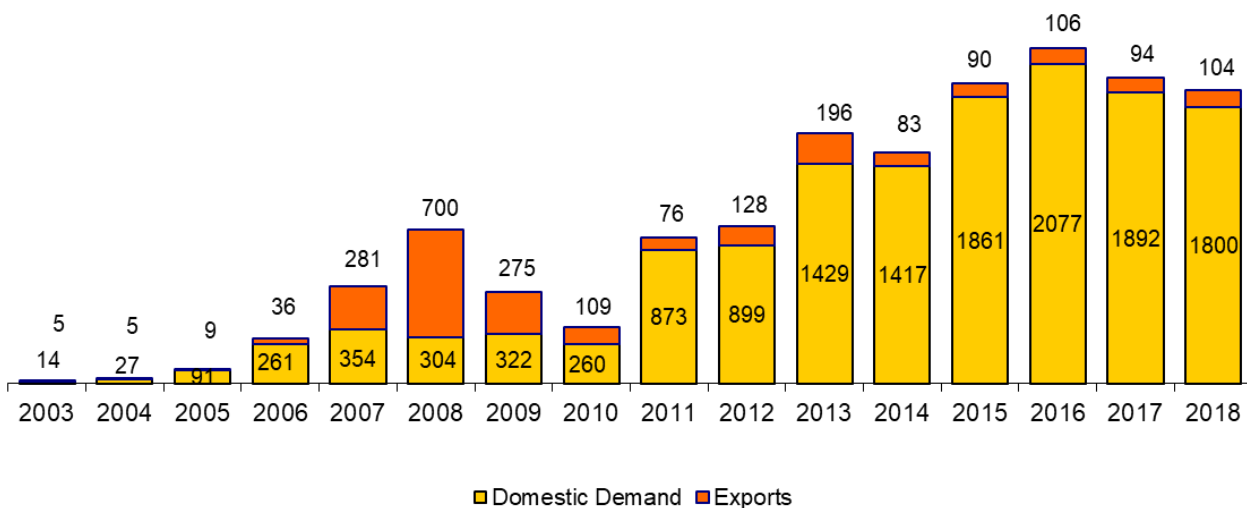


Figure 3. US Biodiesel Demand (million gallons, 2003-2018) [22]

The production of biodiesel in the United States grew from about 343 million gallons to 1.86 billion gallons from 2010 to 2018. This is a CAGR of 23%. This aligns with the demand in the United States. Demand declined in 2017 and 2018 mainly due to import tariffs imposed on biodiesel imports from Argentina and Indonesia, removing volume from the supply. [20] Overall, the total supply of biodiesel increased from 396 million gallons to 2.2 billion gallons, a CAGR of 24% during the 2010 to 2018 period. See Figure 4. [22] Building the proposed supercritical plant in the United States will help displace foreign producers in the country.

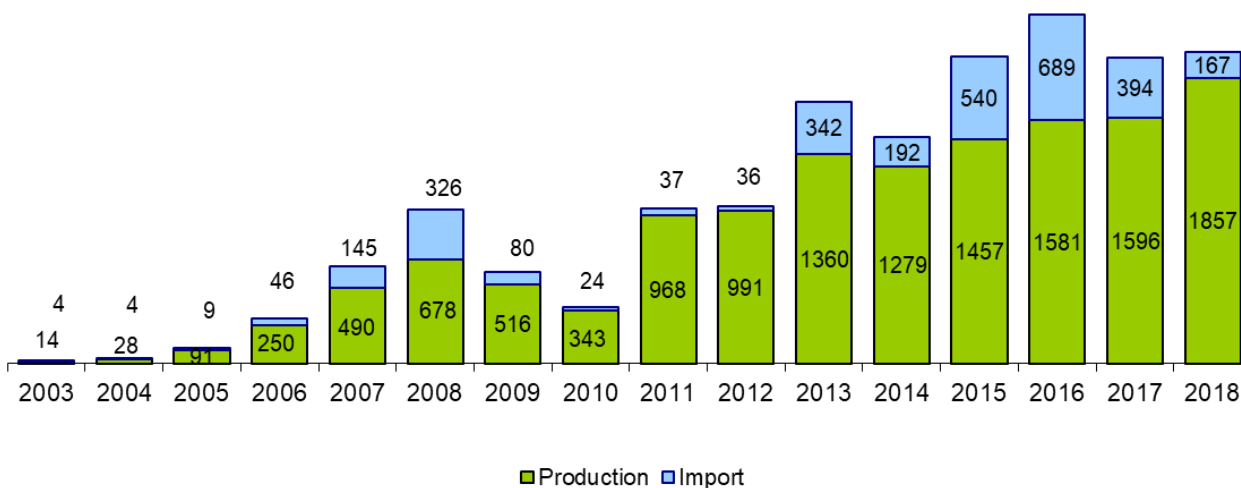


Figure 4. U.S. biodiesel supply, (million gallons 2003-2018) [22]

Figure 5 illustrates that the United States exports of biodiesel has been declining since 2011, indicating that the amount of biodiesel imported exceeded the amount exported. Exports from the United States peaked in 2008 due to the effect of a biodiesel tax credit in the European Union. Exports also declined as production and consumption in the United States increased due to tax credits and the Renewable Fuel Standard. [23]

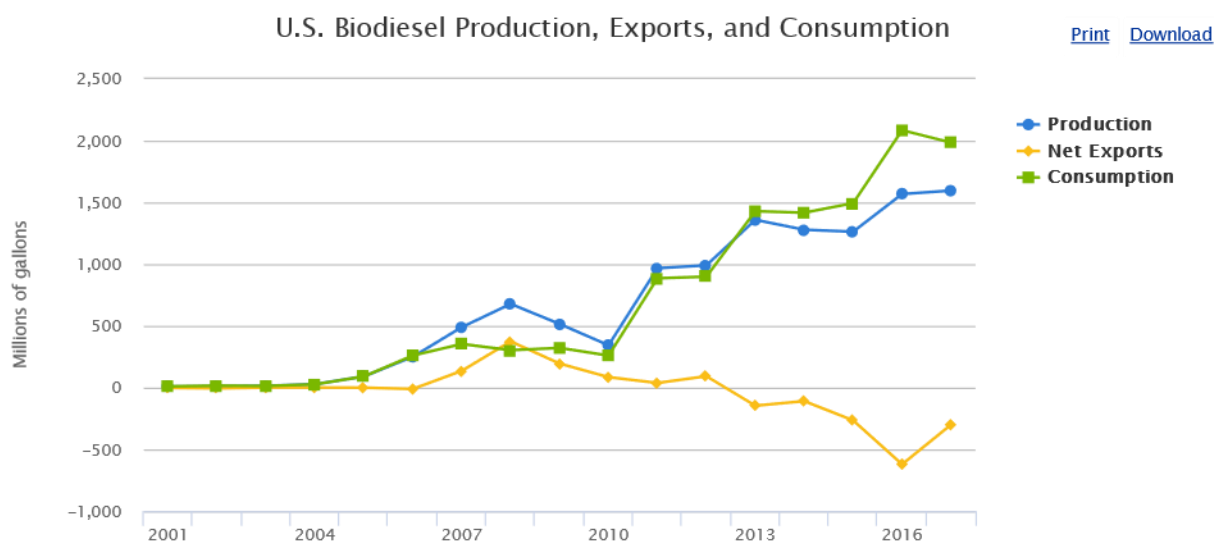


Figure 5. U.S. biodiesel production, exports, and consumption. [23]

The United States has been increasing the number of stations where biodiesel and biodiesel blends can be obtained nationwide. As of 2018, there were 681 stations nationwide, a growth of 38% from 2002 to 2018. [22] Most of the stations are located near highways and concentrated on the west coast and within the Midwest. See Table 3 and Figure 6. [22]

Table 3. Number of alternative fuel stations: 1992-2018 [22]

Year	Biofuels		Others ²	Total
	E-85 (85% ethanol, 15% gasoline)	<u>Biodiesel</u>		
1992	2	0	3,689	3,691
1993	7	0	3,844	3,851
1994	32	0	4,423	4,455
1995	37	0	4,640	4,677
1996	68	0	6,032	6,100
1997	71	0	6,168	6,239
1998	40	0	7,229	7,269
1999	49	0	6,007	6,056
2000	113	2	5,090	5,205
2001	154	16	5,372	5,542
2002	149	79	5,513	5,741
2003	188	142	5,900	6,230
2004	200	176	5,344	5,720
2005	436	304	4,424	5,164
2006	762	459	3,405	4,626
2007	1,208	742	3,601	5,551
2008	1,644	645	3,467	5,756
2009	1,928	679	3,804	6,411
2010	2,151	644	4,117	6,912
2011	2,442	627	7,002	10,071
2012	2,553	675	17,270	20,498
2013	2,639	757	23,763	27,159
2014	2,840	783	30,182	33,805
2015	2,990	721	36,252	39,963
2016	3,091	697	47,594	51,382
2017	3,322	702	56,029	60,053
2018	3,617	681	66,264	70,562

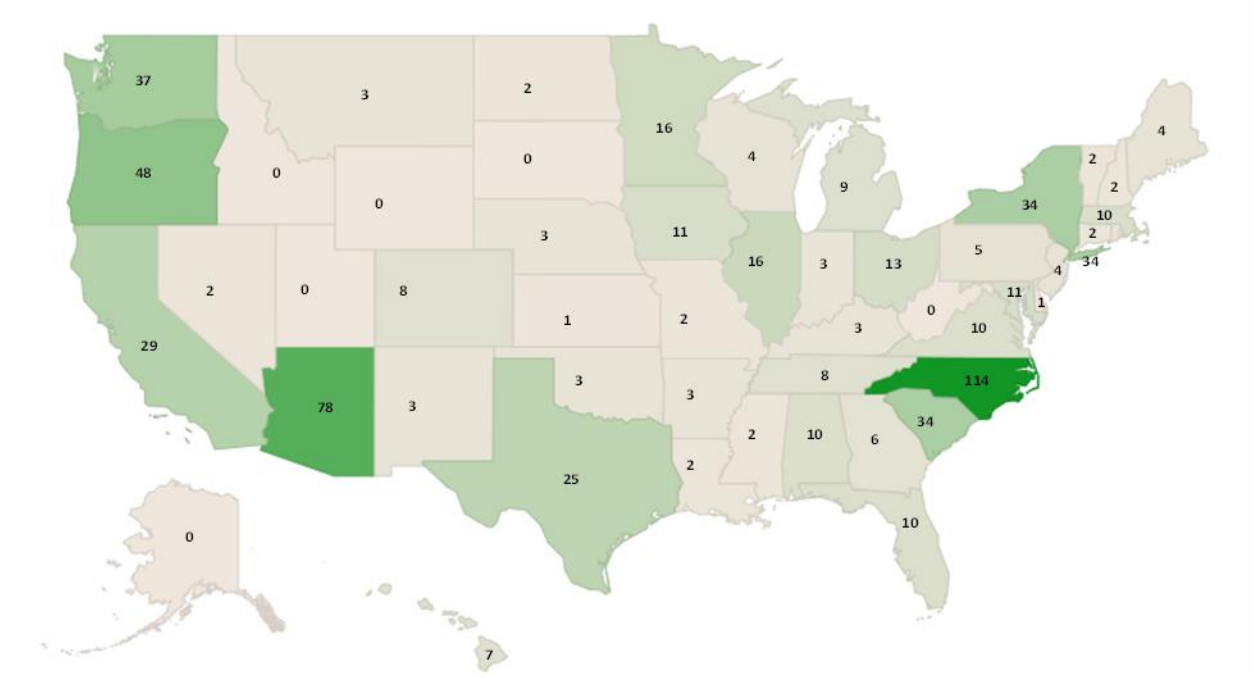


Figure 6. Number of biodiesel fuelling stations per state. [25]

Illinois Market

Illinois is the fourth largest biodiesel producing state, producing 184 million gallons, 7% of the total production amount in the entire United States. It currently has 6 biodiesel plants including a plant using the supercritical process – our process of choice. It also has 16 biodiesel fuelling stations and is surrounded by states that also have multiple fuel stations. See Table 4 and Figure 6. [22] “Illinois is the fifth largest energy-consuming state in the nation, and its industrial sector, which includes petroleum refining and coal mining, uses the most energy of any end-use sector in the state.” [24] It is also the seventh state in producing carbon dioxide emissions, producing 201.0 million metric tons of CO₂ in 2017. [26] In January of 2019, Illinois entered into the U.S. Climate Alliance, a group of states committed to reducing greenhouse gas emissions and has pledged to reduce carbon pollution and to promote clean energy including biodiesel. [27] These policies will impact and increase the demand for biodiesel in Illinois.

Table 4. Biodiesel: Production capacity, by State: December 2019 [22]

Rank	State	Number Plants	Production	
			Million gallons per year	Percentage of Total
	U.S. Total	95	2,537	
1	Iowa	10	445	18%
2	Texas	9	377	15%
3	Missouri	9	246	10%
4	Illinois	6	184	7%
5	Arkansas	3	115	5%
6	Washington	2	112	4%
7	Indiana	3	107	4%
8	Mississippi	3	102	4%
9	Pennsylvania	2	90	4%
10	North Dakota	1	85	3%
11	Minnesota	3	85	3%
12	California	9	85	3%
14	Kansas	2	60	2%
15	Kentucky	1	54	2%
16	Nebraska	1	50	2%
17	Tennessee	2	38	1%
18	Oklahoma	1	38	1%
19	Wisconsin	2	32	1%
20	Connecticut	1	30	1%
21	Alabama	1	20	1%
22	Georgia	3	19	1%
23	Michigan	2	18	1%
24	Oregon	1	17	1%
25	Florida	2	13	1%
26	Virginia	4	9	0%
27	North Carolina	2	7	0%
28	Rhode Island	1	7	0%
29	Hawaii	1	6	0%
30	South Carolina	1	5	0%
31	New Hampshire	1	4	0%
32	Arizona	1	2	0%
33	New Mexico	1	1	0%
34	Massachusetts	1	1	0%
35	Maine	1	1	0%
36	Alaska	1	0	0%

By-product:

Glycerin is a by-product of the transesterification reaction. It is a trihydroxy sugar alcohol with three carbon atoms and three hydroxyl groups. [28] It is used in various industries including the food and beverage, alkyd resins, cosmetic, industrial chemicals, pharmaceutical and tobacco industries. [29] See Figure 7.

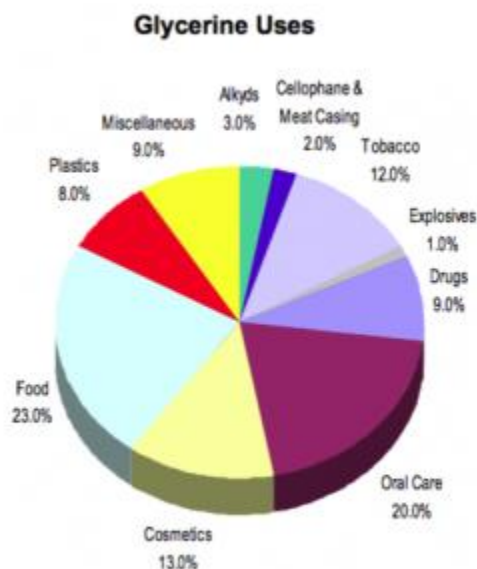


Figure 7. Uses for glycerine [29]

Glycerin can be categorized as crude, technical grade, and USP grade. Crude glycerin contains 40 to 88% glycerin with the rest impurities such as methanol, water, soaps, and salts. Technical grade glycerine is refined and is 98% pure minimum. It contains no methanol, soaps, and salts. USP grade glycerine is pharmaceutical grade and is suitable for food, cosmetics, pharmaceuticals and other applications and is 99.7% pure. [30] Currently the glycerin market is driven by the personal care, cosmetic, and pharmaceutical industries, thus making technical grade glycerine in high demand. The use of glycerine as a substitute for antifreeze (propylene and ethylene glycols) is expected to improve market growth. [31]

As of 2018, the global glycerin market was valued at \$2.998.4 million USD with an estimated growth of 5.1% CAGR from 2018 to 2028. East Asia and South Asia dominate the market in terms of consumption and demand. Low production costs, labor costs, and lenient laws and regulations are causing manufacturers to set up facilities in these regions. In Europe and North America, slow growth in the glycerin market is expected. [31] The major countries producing glycerine include the European Union at 25%, China at 20%, and the United States at 15%. The major countries consuming glycerine include China at 25%, Western Europe at 22%, the United States at 17%, and the rest of Europe at 9%. [32]

Due to the increase in production of biodiesel around the globe, the production of glycerine has increased. The United States Department of Energy reported that in 2017, the global biodiesel consumption was around 35.7 million tons which caused a 2.3% year-over-year growth in glycerine production. [31] In 2016 and 2017, the global supply of glycerin was 4,021 thousand metric tons and was larger than the consumption which was 3,125 thousand metric tons. See Figure 8 and Figure 9. [32] [33]

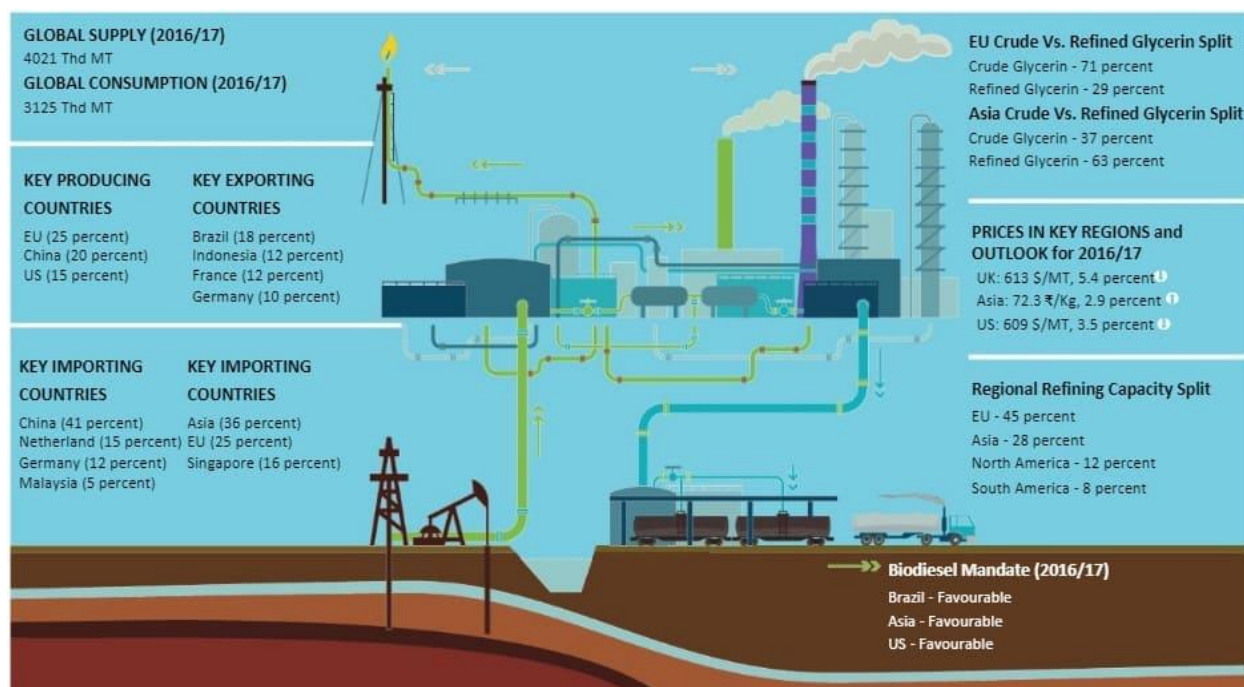


Figure 8. Market statistics for glycerin in 2016/2017 [32]

Glycerin Market - Growth Rate by Region, 2019-2024



Figure 9. Market growth by region 2019-2027 [33]

The market is highly concentrated with five major companies dominating about 65% of the total demand. [32] Figure 10 shows pricing trends for key producing countries from 2019 through mid-2020. Prices in the United States are expected to drop 2 cents per pound by mid-2020. [34] These prices are expected to remain low in the foreseeable future due to the extension of the tax credit for biodiesel production. [34]

PRODUCT	USA	Europe	China	Forecast	Est. for USA	Est. for Europe	Est. for China
	Delivered in bulk December 2019	Delivered in bulk December 2019	CIF in bulk in flexibags December 2019		Mid 2020	Mid 2020	Mid 2020
99.7% kosher quality refined glycerine spot price	37 c/lb.	EUR 540 pmt	USD 450 pmt	→	35 c/lb.	EUR 530 pmt	USD 430 pmt
99.5% technical quality refined glycerine spot price	30 c/lb.	EUR 475 pmt	USD 375 pmt		28 c/lb.	EUR 450 pmt	USD 350 pmt
80% crude glycerine for refining spot price	15 c/lb.	EUR 330 pmt	USD 170 pmt		12 c/lb.	EUR 300 pmt	USD 165 pmt

Figure 10. Glycerin prices for USA, Europe, and China through mid-2020. [34]

Feedstock:

Biodiesel can be produced using many different feedstocks including soybean oil, canola oil, animal fats, corn, and recycled cooking oil. See Figure 11. [35]

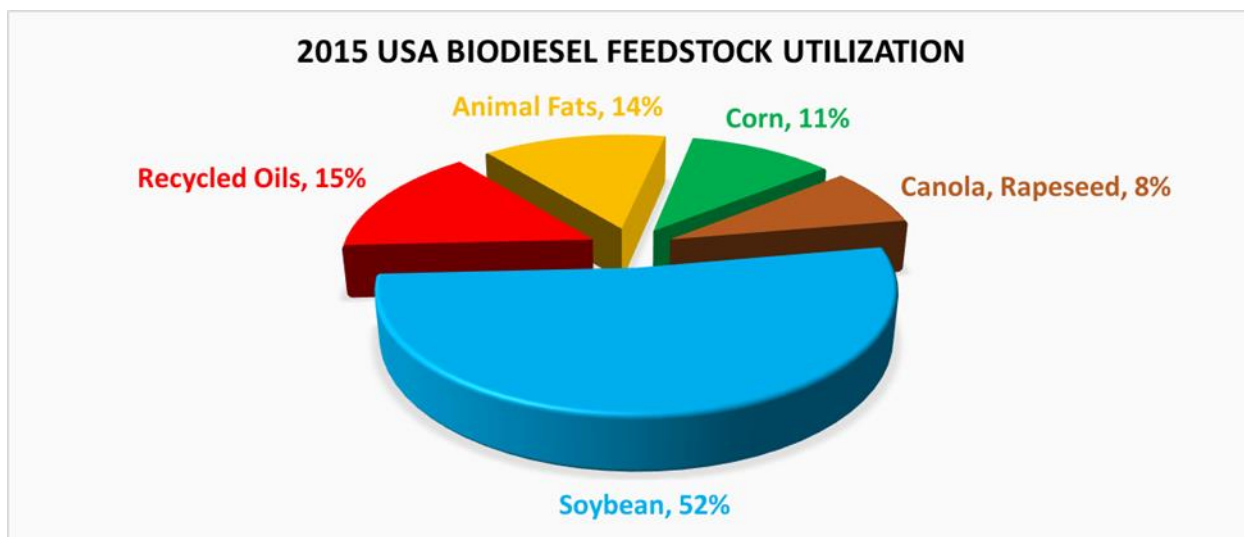


Figure 11. U.S. biodiesel feedstock 2015 [35]

Biodiesel production cost is very sensitive to changes in feedstock prices and can significantly affect profitability as feedstock accounts for 73% of the cost of production. See Figure 12. [36]

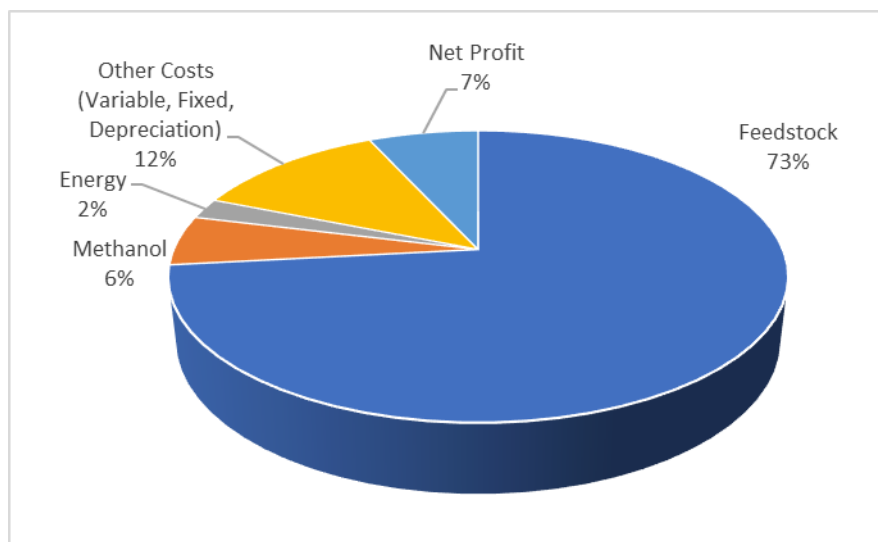


Figure 12. Biodiesel Cost Structure (% per gallon, 2018) [36]

For example, in 2016-2018, feedstock prices increased 7.7% as prices for soybean oil increased sharply and at the same time, the market price for biodiesel fell sharply, making it difficult to break even without tax incentives and tax credits. [37]

Figure 13 illustrates the biodiesel pricing trends from 2000 to 2019. [38]

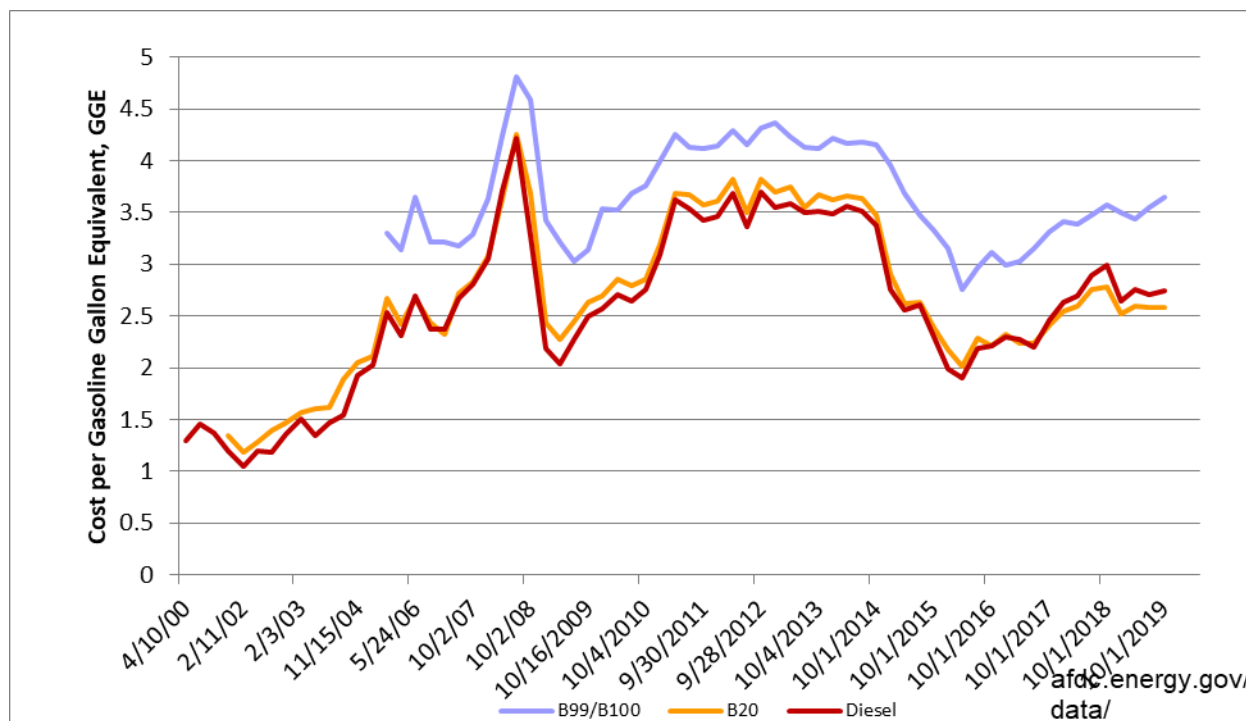


Figure 13. Biodiesel pricing trends [38]

The feedstock for the proposed supercritical biodiesel plant is used cooking oil and methanol.

Used Cooking Oil Market Overview:

Used cooking oil (yellow and brown grease) is collected by recycling companies from restaurants and other foodservice businesses to sell to companies that produce biodiesel. [39] “According to the National Renderers Association, in 2018, approximately 4.4 billion pounds of used cooking oil is collected annually from restaurants, grocers, hotels and casinos, and other foodservice providers in the US and Canada”. [40]

By 2024, the recycled cooking oil industry is expected to grow by 1.7% each year until 2024 to reach 3 billion dollars in revenue in the United States. This is driven by the increased demand in biodiesel production and the governmental incentives to lower greenhouse gas emissions. [39] The government is also expected to put more restrictions on how used oil is disposed of, making recycling necessary, and possibly increasing the supply. [39] See Figure 14.

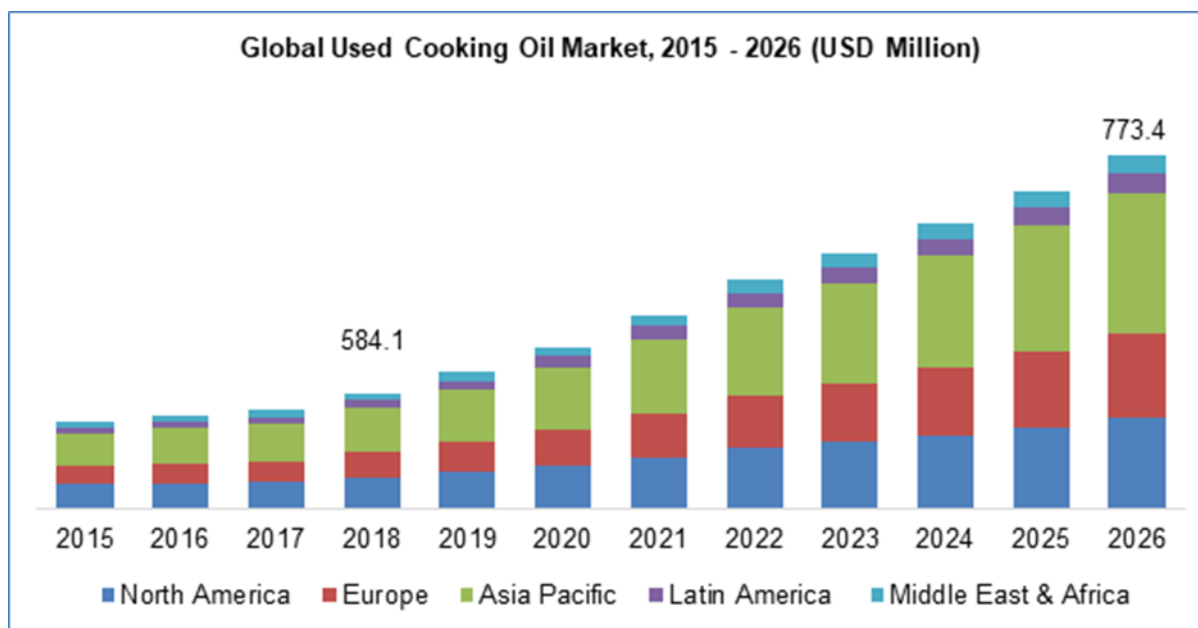


Figure 14. Global used cooking oil market. [41]

Figure 15 describes the pricing trends of used cooking oil in Euros per ton and USD per ton from January 2017 to September 2019. The ending price of used cooking oil as of September 19 is \$705 USD per ton. [42] The declining price from 2017 to 2018 is consistent with the decrease in demand and due to a decrease in production of biodiesel caused by the discontinuation of tax credit incentives for that year in the United States.

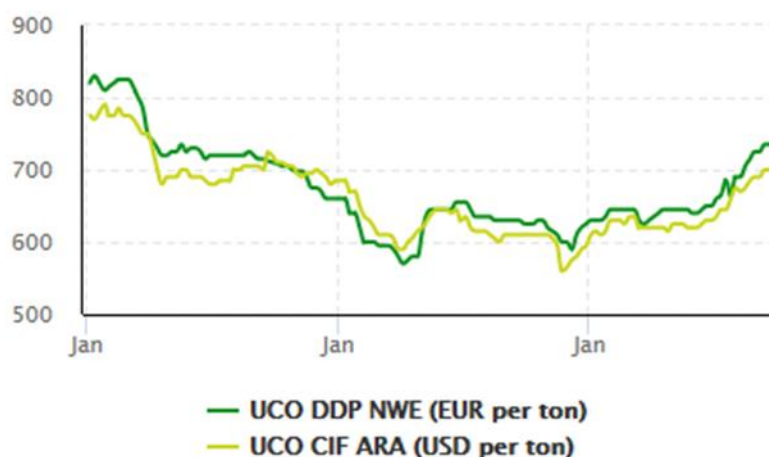


Figure 15. Price of used cooking oil trends from 2017 to 2019 in USD per ton and EUR per ton [42]

Methanol Market Overview:

Methanol is used for the production of many chemicals including olefins, adhesives, biodiesel, formaldehyde, and other biofuels. The global market was estimated at 31.81 billion dollars in 2018 and has an estimated CAGR of 2.9% from 2019 to 2025. This is a 45% growth over the past 5 years. This growth is propelled by the increase production and consumption of formaldehyde and biodiesel. [43] Top five methanol producing countries include China, Saudi Arabia, Trinidad and Tobago, Iran, and Russia. Globally, 36.6 billion gallons are produced by over 90 plants meeting the global demand. The United States, with the development of shale gas, has decreased their reliance on imported methanol and dropped the price of natural gas 65%. [44] Natural gas is used as feedstock in the production of methanol. The overall production of methanol depends greatly on the costs of the feedstock, which makes up approximately 90% of the total cost of production. In North America, including the United States, the methanol market is the second largest globally. The demand for methanol is mainly driven by the construction industry that uses large amounts of formaldehyde. [41] The price of methanol in the United States was about \$326 USD per metric ton in 2019. See Figure 16.

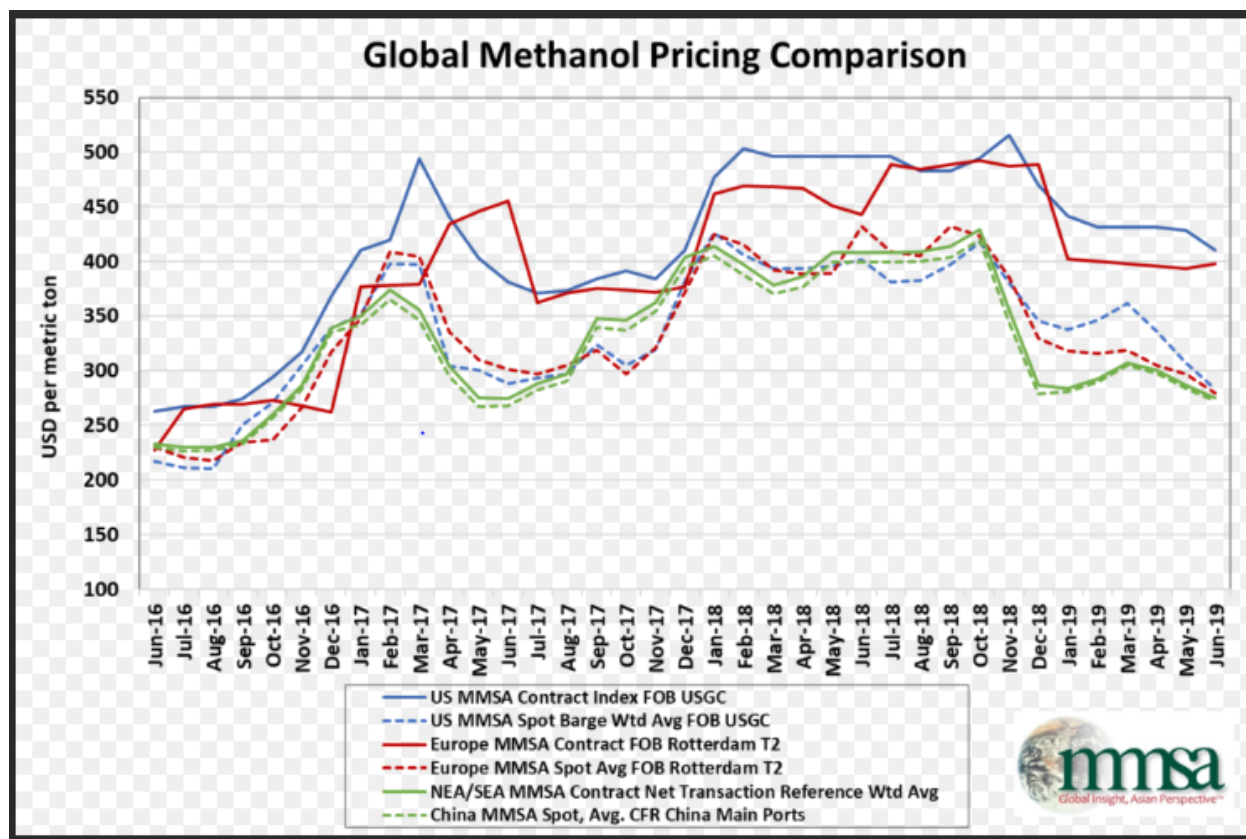


Figure 16. Global methanol price trends 2016-2019 [45]

Summary:

The main concern for biodiesel production and profitability are feedstock prices and availability as well as an unpredictable market. The success of biodiesel production is largely dependent of tax credits and incentives, which have been historically extended each year.

4. Process Description**4.1. Current Methods of Production****4.1.1. Catalytic Process**

The catalytic process itself relies on an acid or base type catalyst to push the transesterification reaction forward. The types of base catalysts that can be used would be either methanol or ethanol. [48,49] The types of acid catalysts that can be used would be sulfuric acid or hydrochloric acid. [47] Each of the catalysts whether they be acid or base come with their own byproducts that would then be needed to be further cleaned from the final biodiesel product. The catalytic process could be either batch or continuous.

The catalytic processes operate under a range of pressures up to 5bar and temperatures of up to 140°F. [46] The benefit of this type of production process is that with such relatively stable temperatures and pressures it can be operated year-round regardless of whether it's winter or summer. Another benefit of the catalytic process is that the raw materials are relatively inexpensive and not a lot of methanol or sulfuric acid is needed for the transesterification reaction to take place. The base catalytic process finds itself with high yields around 98.5%. [50]

The drawback to the catalytic process regardless of whether an acid or base catalyst is used is that further cleaning is needed to purify the final product. Base catalytic processes cause saponification in the product and would need to be cleaned along with the forming glycerol's. While acid catalysts avoid the issue of saponification, equipment would need to be regularly checked in order to avoid corrosion. [47] Corrosive equipment means the material of construction will need to be of more durable type steels that will also need to be changed regularly. The resulting corrosion side effect brings with it a higher capital cost. The acid catalysts find it's yields are 90% and produced more undesirable side reactions than the base catalyst method. [51] Both catalytic methods find themselves purchase more equipment for the sole purpose of cleaning the final product because of undesirable byproducts or side reactions that form.

4.1.2. Enzyme Catalyst Process

The enzyme process doesn't additive methanol or acid to push the reaction forward. The algae do all the work to create the fatty raw material. The most common enzyme currently in industry would be Lipase. [53] The temperatures and pressures for the enzyme catalytic process are in the same range as the catalytic process. While the enzyme catalytic process can be continuous, it is a batch process most of the time because of the need to regenerate the enzyme.

A big benefit that the enzyme catalyst has over all other competing processes is that there are no byproducts or side reactions formed. The enzyme catalyst takes living algae with water and is highly insensitive to the FFA. This is because it doesn't produce unwanted CO₂, NO_x, or sulfides. [47] The enzyme catalytic process is the preferred greener process since it takes CO₂ when causing the process to be pushed forward.

The enzyme catalytic process comes with its own set of cons. One of the issues when trying to run the enzyme catalytic process is that the enzymes require extraordinary volumes of water for the living algae. [47] The algae are also very sensitive to the alcohol content in the process. The overall standard yields are about 96% which is higher than the acid-catalyst process but lower than the base-catalyst process. [52] The enzymes tend to be more expensive as well than the standard base catalysts or acid catalysts when running the process. The catalyst also must be replaced periodically thereby increasing expenses.

4.1.3. Non-Catalytic Supercritical Process

In the case of the supercritical process, it differs greatly from any other method of biodiesel production in the fact that a catalyst is not needed. By pushing the oil, regardless of the concentration of free fatty acid with a mixture of methanol to its supercritical state, it will cause the transesterification reaction to proceed. The supercritical process itself can be either in a batch or continuous state. While other catalytic based processes operate at relatively similar temperature and pressure conditions, the supercritical process find itself at about 752°C and 2901 psig. [54]

The advantage that the supercritical process has over its competitors is two-fold. The supercritical process produces the highest yields and conversion out of all the methods. While the base catalytic method can reach yields up to 98%, the supercritical can reach yields up to 99% and above. In certain cases, a 100% yield was observed. The other benefit of a supercritical process is that the reaction time is substantially less than its counterparts. While the time for the transesterification reaction for the other methods would require about a 2+ hour reaction time, the supercritical process requires a reaction time of about 20 minutes at worst. [54] This means that more biodiesel can be produced at a faster rate than others.

On the other hand, the supercritical process operates at temperatures and pressures which are difficult to control and require large costs for equipment that could withstand such fierce operating conditions. The ratio of alcohol to oil for this process is another factor that works against it. Since there is no catalyst involved, more methanol is needed at the supercritical state to push the reaction to completion. While a significant volume of methanol would be recycled back once the reaction is completed, there would still need to be that initial supply of methanol for the first reaction to take place. About a 40:1 ratio of methanol to oil is needed. [54]

4.2. Our Chosen Method of Production

The supercritical method for producing biodiesel was ultimately chosen as our solution because of the numerous benefits it provides. As stated above, it offers high yields of biodiesel with high purity and is able to do so much faster than the traditional catalytic method. An added benefit to using this process is that the glycerol, a by-product of the reaction, is of higher quality than when produced through other methods. The glycerol produced through the supercritical process is technical grade and is worth approximately double the standard selling price of glycerol. [56, 60]

Another key aspect of this process that makes it ideal for biodiesel production is its ability to process feedstocks with up to 100% free fatty acids (FFA). [56] The robustness of this process means it can effectively process used cooking oil without need to reduce the amount of FFAs in the feedstock prior to reacting. Since used cooking oil will be used as a feed stock, our process offers a safe and more environmentally friendly method for disposing of this used oil. It also eliminates the need to remove a potential food source from the market since the oil has already been used and is considered waste.

Lastly, the supercritical production process allows biodiesel to be produced in a streamlined process. The traditional catalytic method requires extra equipment to separate out the catalyst used in the transesterification reaction. In fact, the separation of the catalyst typically requires in extra reactor, heated tank, and distillation column to effectively separate it from the glycerol by-product. This equipment adds to the capital cost of the plant, as well as adding to the cost of the utilities during operation. The supercritical process eliminates three costly steps in the tradition process and can operate at up to 38% lower production cost. [56]

4.3. Detailed Process Description

In general, the production of biodiesel is accomplished by reacting an oil with an alcohol. Typical processes use vegetable oils as the feedstock and react them with methanol or ethanol to produce fatty acid methyl esters (FAME) and glycerol. The chemistry is shown in Figure 17 below.

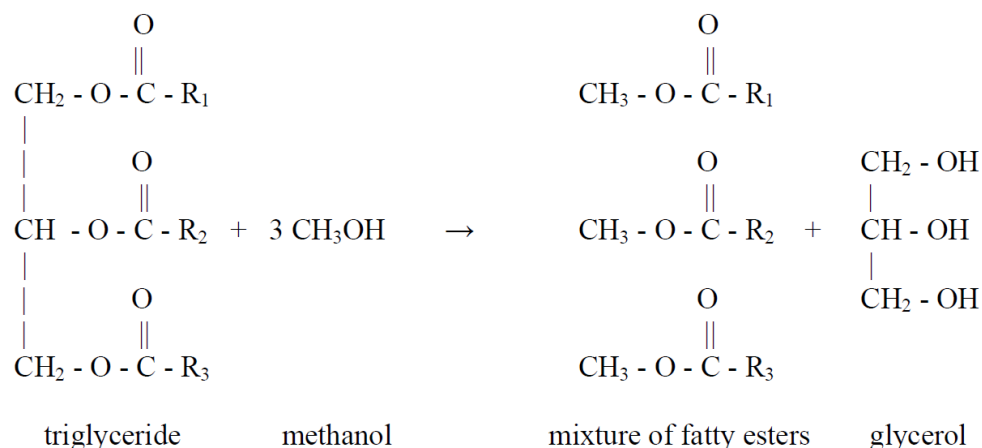


Figure 17. General reaction chemistry for biodiesel production [55]

The main steps in the typical processing of biodiesel includes preprocessing, transesterification reaction, methanol recovery, and separation. The details of these steps including operating conditions and any relevant information is discussed below. A BFD of the main steps in the supercritical process is also shown below in Figure 18.

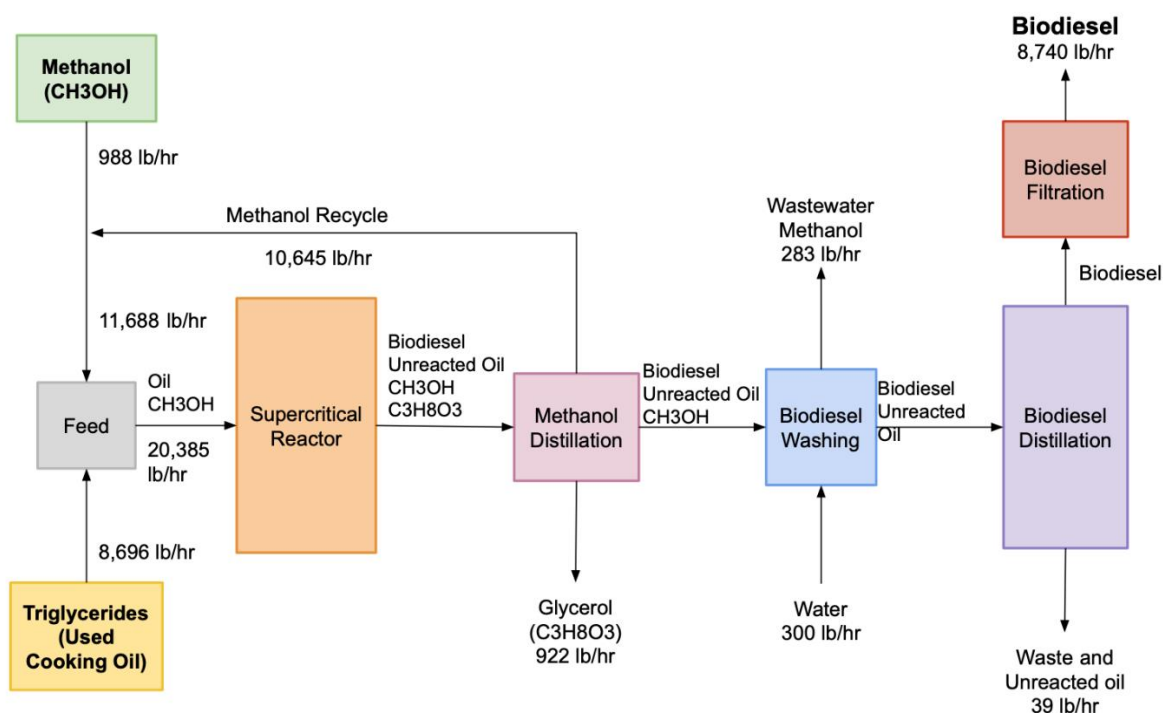


Figure 18. BFD summary of the supercritical production process

Preprocessing:

Preprocessing of the feedstock used cooking oil typically requires simple filtration and heating to remove any impurities in the oil and to ensure all oil is homogeneously mixed. The filtration system that can be used for this include cloth filters, cartridge filters, and rotary vacuum filters. Since the main component that needs to be filtered out of the used cooking oil is the sediment left from the food, the vacuum filtration system would serve as the best method because the waste product is mostly solid, which can be easily disposed of. The prepped oil and methanol are then mixed in a ratio of 37:1 methanol to oil and sent to be preheated. [57] The mixed cooking oil and methanol feed get are pumped through a heat exchanger to preheat the fluid mixture before it flows to the reactor. The heat exchanger chosen for this process is a shell and tube type heat exchanger with high-pressure steam to heat it to around 482°F. This temperature is the required critical temperature for the transesterification reaction to take place. Other types of heat exchangers could have been used such as a double pipe or a plate and fin, but the shell and tube is the standard heat exchanger used in industry. Heat transfer equipment such as a furnace was disregarded in the decision making since the temperature range for our process fluid can be met by high pressure steam.

Transesterification Reaction:

After the used cooking oil has been filtered and preheated, it is sent to the transesterification reactor along with methanol where it undergoes the transesterification reaction. The reaction can take place in either a plug flow reactor (PFR) or a continuous stirred tank reactor (CSTR). The PFR can operate continuously and does not require high pressure seals around an agitator; however, it does not provide good quality control. The CSTR can operate continuously if multiple reactors are used in parallel but requires high pressure seals around the shaft. The reactor type chosen for this process is CSTR reactors in parallel because if a bad batch of biodiesel is produced it can easily be identified and disposed of. The reaction takes place with the chemistry shown above in Figure 17. The reactor conditions to successfully react under supercritical conditions can range for temperature from about 482°F to 1292°F and for pressure about 2466 to 2901 psig. [56, 57] Reaction kinetic data for the transesterification reaction were studied for temperatures up to 1200°F. The data is summarized in Table 5 below.

Table 5. Reaction kinetic and thermodynamic data summary. [57]

Parameter	Value
Reaction Time (min)	12-20
Activation Energy (MBTU/lbmol)	24.0757
Arrhenius Constant (s^{-1})	283.592
[Oil] Reaction Rate Exponent	1.2411
[Methanol] Reaction Rate Exponent	0.1476

To make this process continuous, multiple CSTR reactors will be used in parallel. Two reactors will be needed for the purpose of using one to react for the required time while the other is emptied and filled with new reactants. This will also allow for maintenance on a reactor while the other can continue to run.

Methanol Recovery:

The recovery of methanol takes place in a two-step process consisting of a flash column and in a vacuum distillation column. The first flash column operates at about 158°F and 14.5 psig. The second column, the vacuum distillation column, operates at 320°F and 14.5 psig. A two-step separation process is used in this case because the flash column is able to separate 88.1% of the methanol, but due to the large amount of methanol needed for this process it is desired to have a higher recovery. Adding the second distillation column allows for 99.4% recovery of methanol in total, which is sent as a recycle stream back to the feed methanol. [57]

Washing and Separation:

Separation of the biodiesel from the glycerol and unreacted oil is performed in a decanter and a vacuum distillation column. The decanter serves to remove glycerol and operates at 77°F and 14.5 psig. The biodiesel is then washed using water at about 158°F to remove any trace amounts of glycerol and methanol. The vacuum distillation column further separates the biodiesel from the unreacted oil and any other amount of glycerol. This unit operates at 417°F and 14.5 psig. [61]

Appendix I (Design Basis)

The design scale for this process was chosen from the limiting component. This was taken to be the used cooking oil available for purchase for this project. According to the National Renderers Association, there are about 4.4 billion pounds of used cooking oil collected annually in the United States and Canada. [58] This equates to approximately 5.67 million gallons per year. Therefore, the amount of used cooking oil that will be used as a feedstock for this process is 9.18 million gallons per year. This number is reasonable as it is only 1.59% of the available market. The methanol required for initial charge of this process is approximately 39,188 gallons per day, however, of the unreacted methanol, 99.5% of it will be recycled, which will reduce the amount of methanol that will need to be purchased to approximately 3583.2 gallons per day. The plant will be located in the Midwestern United States, but more specifically in Dekalb Illinois. This would provide adequate land for the plant, and adequate availability of used cooking oil because Mahoney Environmental, a major collector of used cooking oil, is located nearby. [59] Lastly, the biodiesel will be produced per ASTM-6751, which calls for the standards shown below in Table 6. [62]



D6751 – 19

Table 6. ASTM-6751 standards for biodiesel production [62]

Property	Test Method ^A	Grade No. 1-B S15	Grade No. 1-B S500	Grade No. 2-B S15	Grade No. 2-B S500
Sulfur, ^B % mass (ppm), max	D5453	0.0015 (15)	0.05 (500)	0.0015 (15)	0.05 (500)
Cold soak filterability, seconds, max	D7501	200	200	360 ^C	360 ^C
Monoglyceride content, % mass, max	D6584	0.40	0.40	...	...
Requirements for All Grades					
Calcium and Magnesium, combined, ppm (µg/g), max	EN 14538	5	5	5	5
Flash point (closed cup), °C, min	D93	93	93	93	93
Alcohol control					
One of the following shall be met:					
1. Methanol content, mass %, max	EN 14110	0.2	0.2	0.2	0.2
2. Flash point, °C, min	D93	130	130	130	130
Water and sediment, % volume, max	D2709	0.050	0.050	0.050	0.050
Kinematic viscosity, ^D mm ² /s, 40 °C	D445	1.9-6.0	1.9-6.0	1.9-6.0	1.9-6.0
Sulfated ash, % mass, max	D874	0.020	0.020	0.020	0.020
Copper strip corrosion, max	D130	No. 3	No. 3	No. 3	No. 3
Cetane number, min	D613	47	47	47	47
Cloud point, ^E °C	D2500	Report	Report	Report	Report
Carbon residue, ^F % mass, max	D4530	0.050	0.050	0.050	0.050
Acid number, mg KOH/g, max	D664	0.50	0.50	0.50	0.50
Free glycerin, % mass, max	D6584	0.020	0.020	0.020	0.020
Total glycerin, % mass, max	D6584	0.240	0.240	0.240	0.240
Phosphorus content, % mass, max	D4951	0.001	0.001	0.001	0.001
Distillation temperature, Atmospheric equivalent temperature, 90 % recovered, °C, max	D1160	360	360	360	360
Sodium and Potassium, combined, ppm (µg/g), max	EN 14538	5	5	5	5
Oxidation stability, hours, min	EN 15751	3	3	3	3

^A The test methods indicated are the approved referee methods. Other acceptable methods are indicated in 5.1.^B Other sulfur limits may apply in selected areas in the United States and in other countries.^C For additional cold weather considerations, see Appendix X3.^D See X1.3.1. The 6.0 mm²/s upper viscosity limit is higher than petroleum based diesel fuel and should be taken into consideration when blending.^E The cloud point of biodiesel is generally higher than petroleum based diesel fuel and should be taken into consideration when blending.^F Carbon residue shall be run on the 100 % sample (see 5.1.12).

Appendix II (BFD)

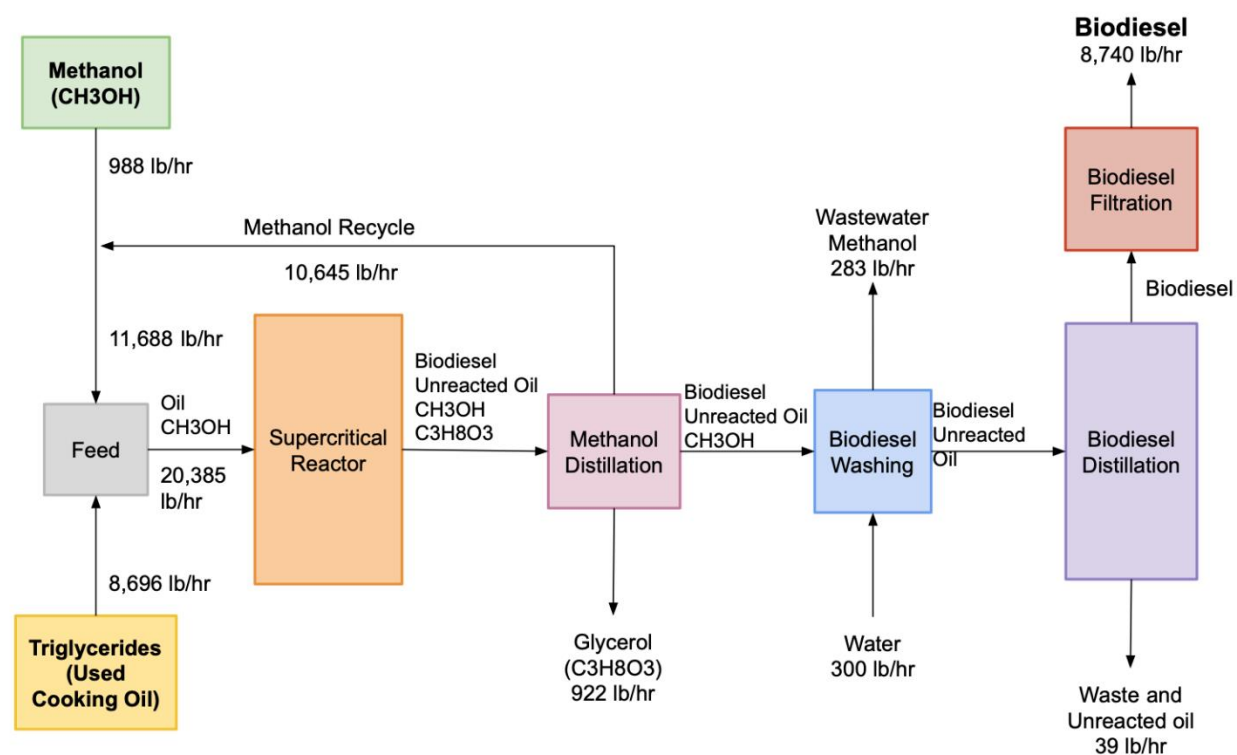


Figure 18. Block flow diagram of the supercritical process for biodiesel production

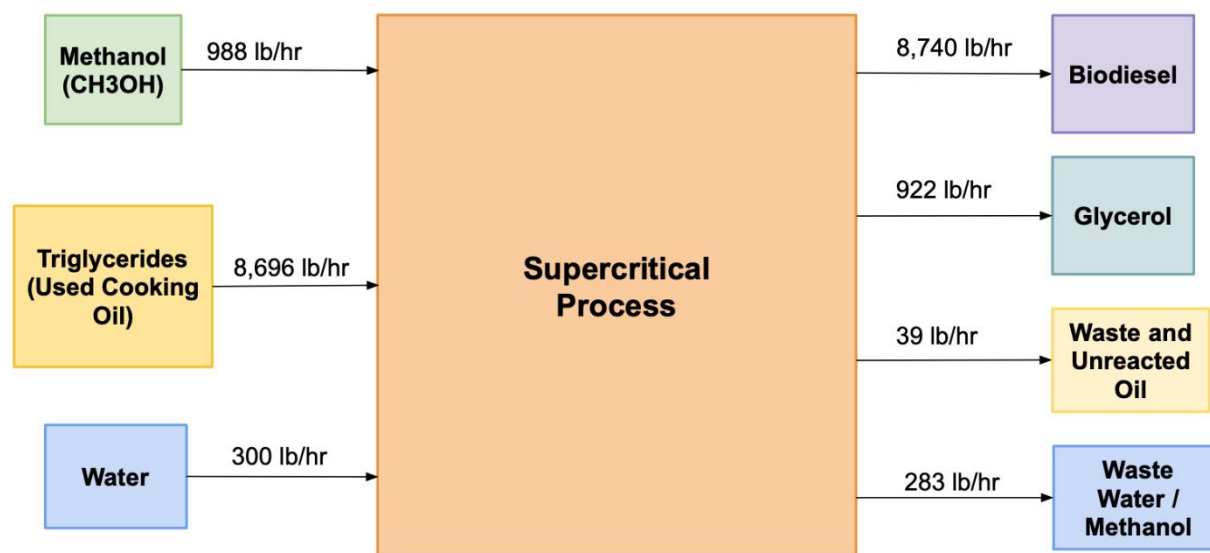


Figure 19. Simple BFD of supercritical process

Table 7. Total mass flow rate balance across the supercritical process

Supercritical Process	Units	In	Out
Total Flow	lb/hr	9,984	9,984

Drawing 1

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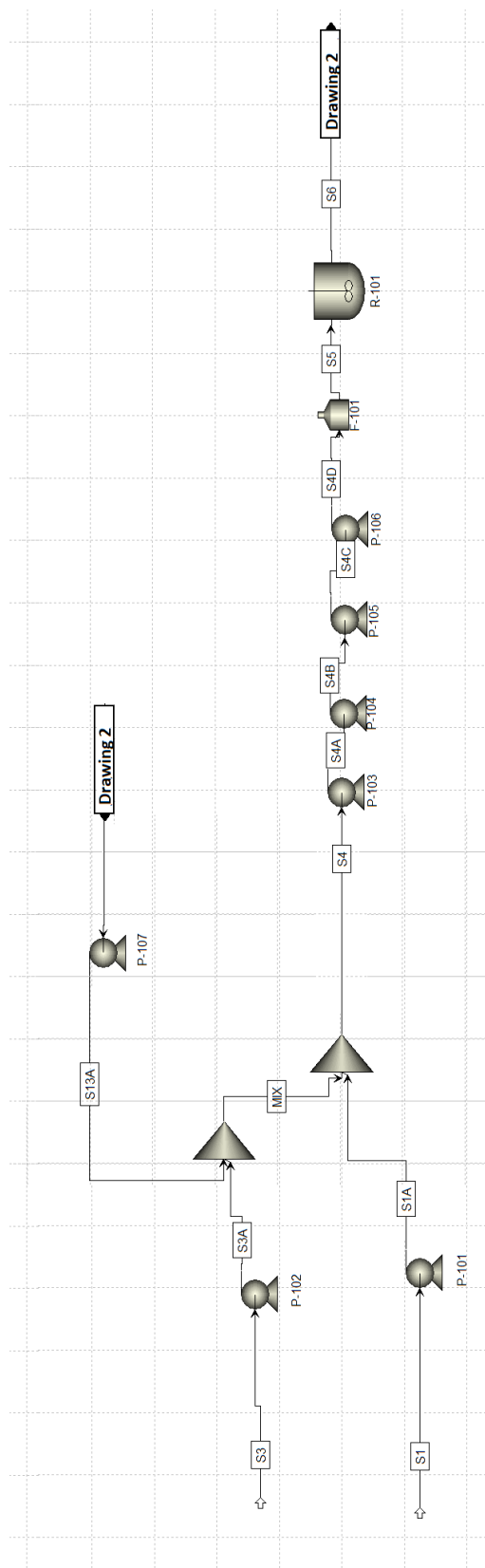


Figure 21. ASPEN Simulation of Drawing 1 (Reaction Section)

Brief explanation of all equipment in PFD Drawing 1:

1. Pumps – A mechanical device that uses suction or pressure to raise or move liquids
 - a. Pump 1 (P-101): Pumps the feed used cooking oil to the transesterification reactor
 - b. Pump 2 (P-102): Pumps the feed methanol to the transesterification reactor
 - c. Pump 3 (P-103): Increases the pressure of the reactor feed
 - d. Pump 4 (P-104): Increases the pressure of the reactor feed
 - e. Pump 5 (P-105): Increases the pressure of the reactor feed
 - f. Pump 6 (P-106): Increases the pressure of the reactor feed
 - g. Pump 7 (P-107): Pumps the recycled methanol back to the reactor feed stream
2. Furnace – Enclosed structure that allows heating of materials to high temperatures
 - a. Furnace (F-101): Used to preheat the reactor feed to the reactors operating condition
3. Reactors – An apparatus or vessel that houses a reaction for the conversion of reactants to products
 - a. Reactor 1 and 2 (R-101a and R-101b): Transesterification reactors (in parallel) that convert feed oil and alcohol to biodiesel at the supercritical state
4. Tanks – Store and materials of a process
 - a. Tank 1 (TK-101): Storage tank for feed used cooking oil
 - b. Tank 2 (TK-102): Storage tank for feed methanol
5. Filters – A device used to remove impurities or solid particles from liquids or gases
 - a. Filter 1 (S-101): A vacuum filter used to remove solid waste particles that may be present in the used cooking oil feed

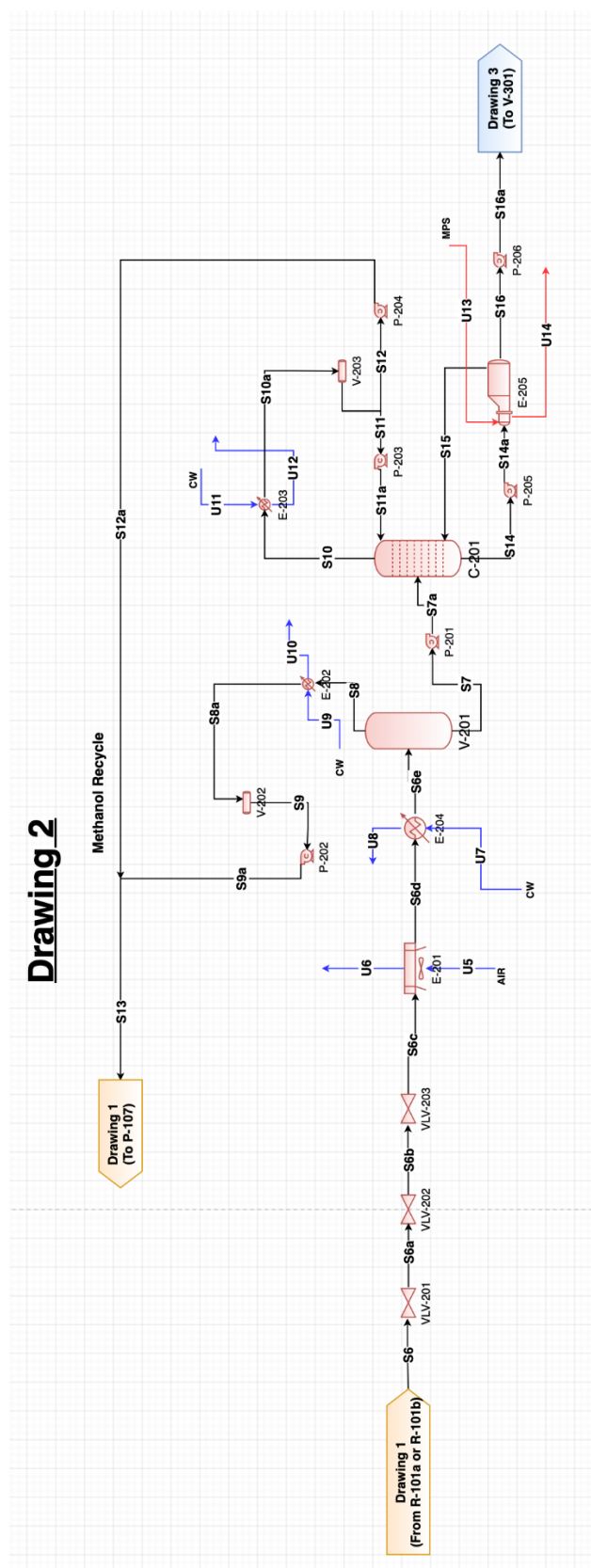


Figure 22. PFD Drawing 2 (Methanol Separation Section)



Brief explanation of all equipment in PFD Drawing 2:

1. Throttle valves – a mechanical device that regulates the supply and lowers pressure of a fluid
 - a. Throttle valve 1 (VLV-201): Lowers the pressure of the reactor effluent
 - b. Throttle valve 2 (VLV-202): Lowers the pressure of the reactor effluent
 - c. Throttle valve 3 (VLV-203): Lowers the pressure of the reactor effluent
2. Pumps – A mechanical device that uses suction or pressure to raise or move liquids
 - a. Pump 1 (P-201): Pumps bottoms product from V-201 to C-201
 - b. Pump 2 (P-202): Pumps recycled methanol from flash column to feed methanol
 - c. Pump 3 (P-203): Reflux pump back into C-201
 - d. Pump 4 (P-204): Top product pump from C-201
 - e. Pump 5 (P-205): Bottoms pump of C-201 into the reboiler
 - f. Pump 6 (P-206): Bottoms product pump from C-201
3. Heat exchangers – System used to transfer heat between two or more fluids
 - a. Heat Exchanger 1 (E-201): Air cooler used to cool reactor effluent
 - b. Heat Exchanger 2 (E-202): Condenser used cool the methanol that is removed from the flash column
 - c. Heat Exchanger 3 (E-203): Condenses the distillate from C-201
 - d. Heat Exchanger 4 (E-204): Cools the reactor effluent further before V-201
 - e. Heat Exchanger 5 (E-205): Reboiler for C-201
4. Columns – A tall metal cylinder internally fitted with perforated horizontal plates or fillings to promote phase separation
 - a. Trayed Distillation Column 1 (C-201): Separates unreacted methanol from transesterification reactor product
5. Vessels – Store and dispense temperature sensitive materials
 - a. Flash Vessel (V-201): A vertical vessel that recycles most of the unreacted methanol from the biodiesel
 - b. Reflux Drum (V-202): A horizontal vessel that temporarily stores distillate from V-201 before sending to feed methanol stream
 - c. Reflux Drum (V-203): A horizontal vessel that temporarily stores distillate from C-201

Drawing 3

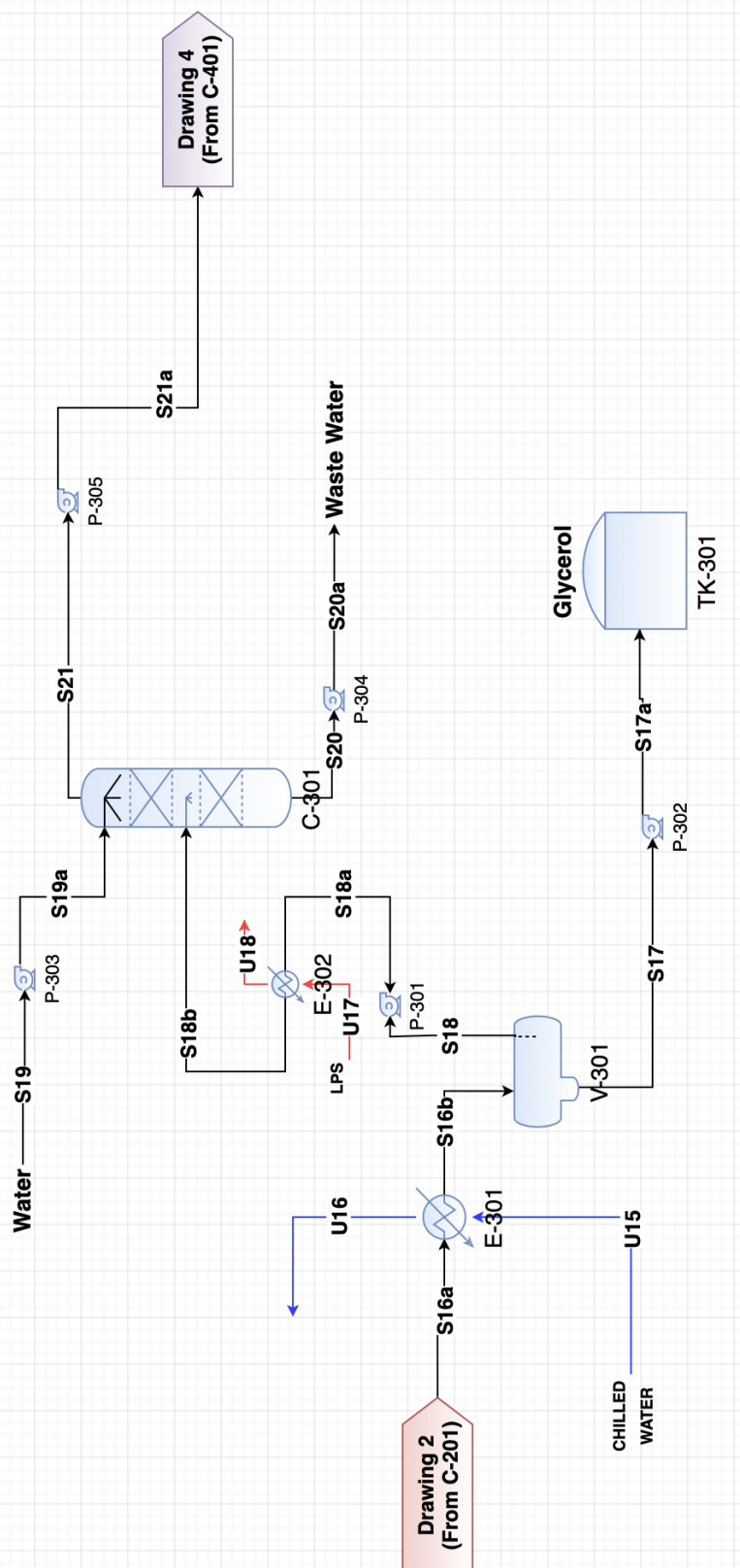
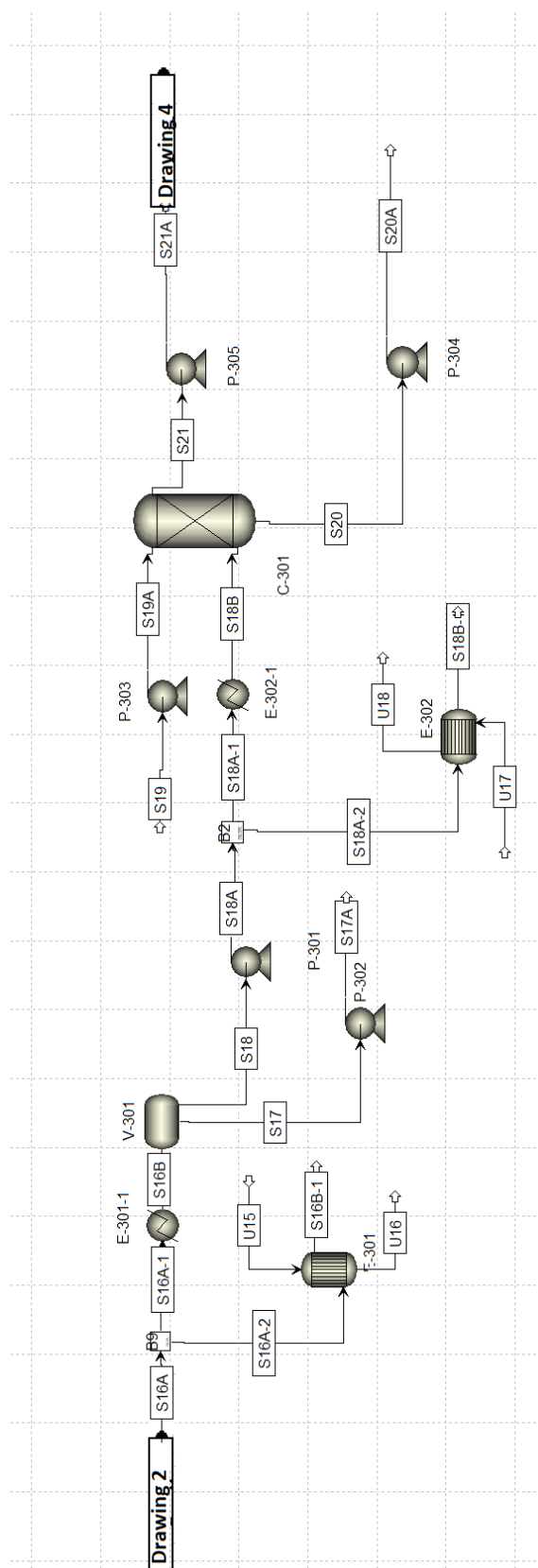


Figure 24. PFD Drawing 3 (Washing and Glycerol Separation Section)



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Brief explanation of all equipment in PFD Drawing 3:

1. Pumps – A mechanical device that uses suction or pressure to raise or move liquids
 - a. Pump 1 (P-301): Decanter top product pump
 - b. Pump 2 (P-302): Decanter bottom product (glycerol) pump
 - c. Pump 3 (P-303): Water feed pump to C-301
 - d. Pump 4 (P-304): Wastewater pump from C-301
 - e. Pump 5 (P-305): Top product pump from C-301
2. Heat exchangers – System used to transfer heat between two or more fluids
 - a. Heat Exchanger 1 (E-101): Heat exchanger used to cool feed to decanter
 - b. Heat Exchanger 2 (E-102): Heat exchanger used to heat feed to washing column
3. Columns – A tall metal cylinder internally fitted with perforated horizontal plates or fillings to promote phase separation
 - a. Washing Column (C-301): Washes biodiesel product to remove any excess methanol
4. Vessels – Store and dispense temperature sensitive materials
 - a. Decanter (V-301): A sediment free vessel that separates out the glycerol from the biodiesel and unreacted oils
5. Tanks – Store and materials of a process
 - a. Tank 1 (TK-301): Storage tank for glycerol product

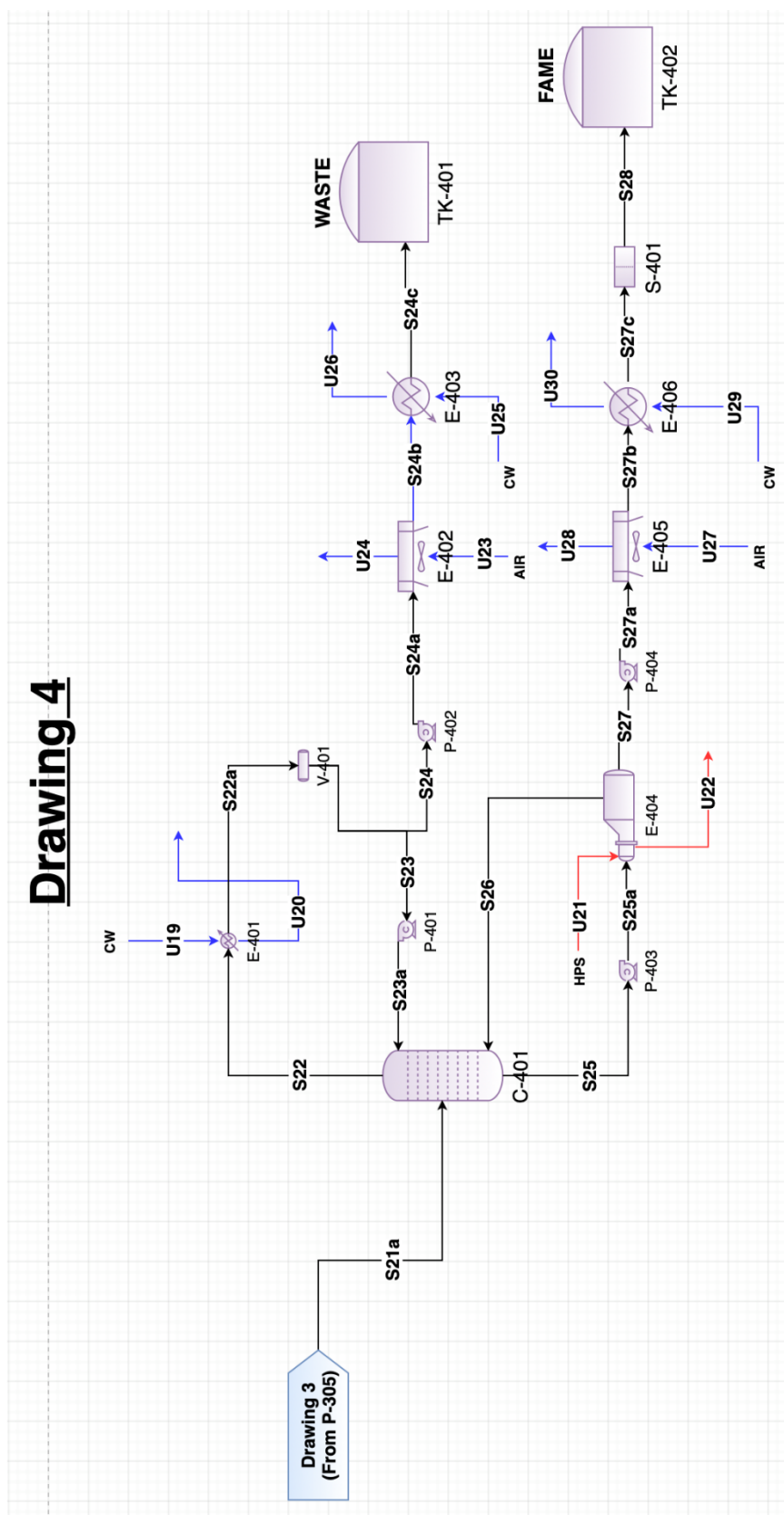


Figure 26. PFD Drawing 4 (Biodiesel Separation Section)

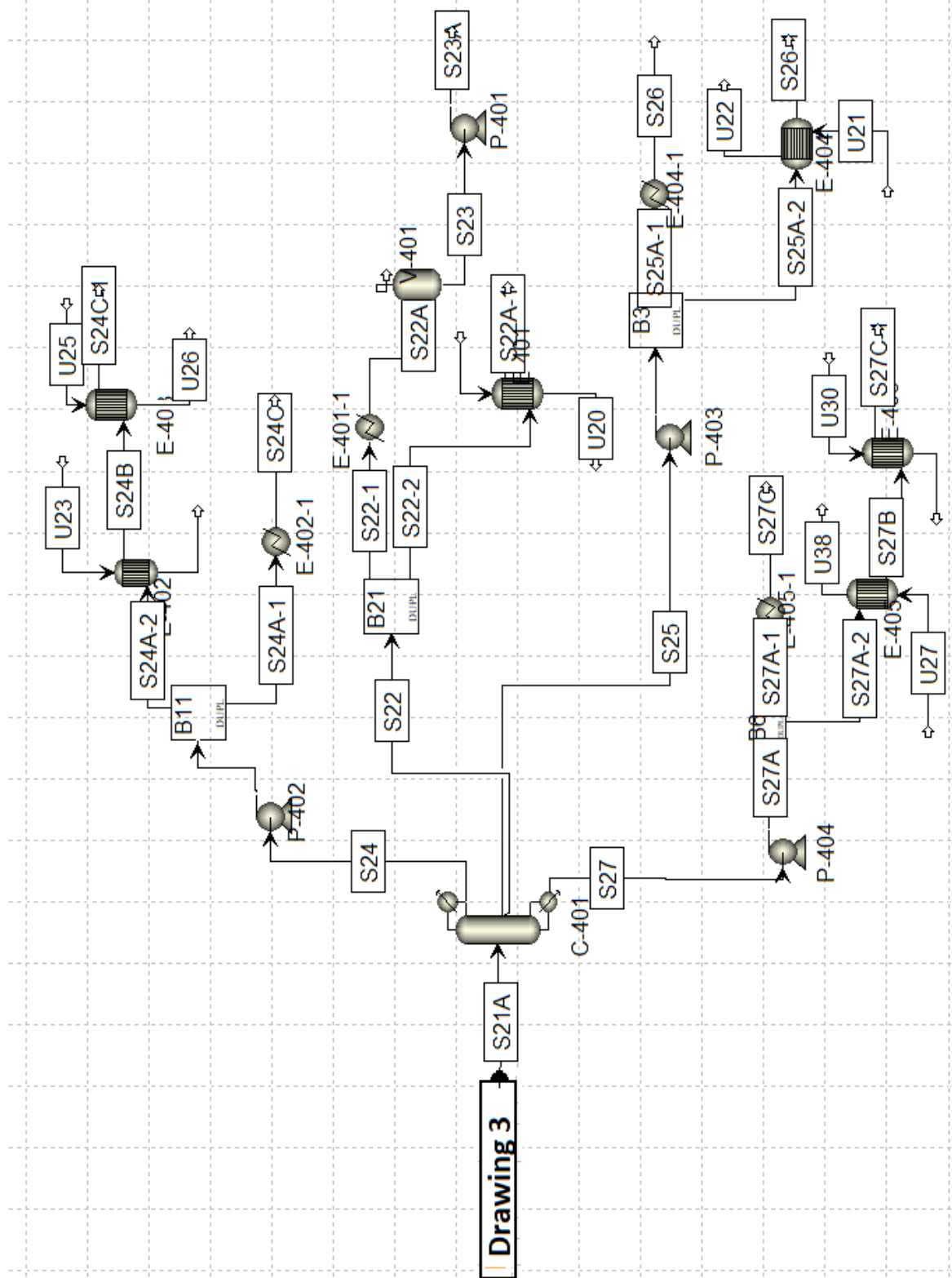


Figure 27. ASPEN Simulation of Drawing 4 (Biodiesel Separation Section)

Brief explanation of all equipment in PFD Drawing 4:

1. Pumps – A mechanical device that uses suction or pressure to raise or move liquids
 - a. Pump 1 (P-401): Reflux pump back into C-401
 - b. Pump 2 (P-402): Top product pump from C-401
 - c. Pump 3 (P-403): Bottoms pump of C-401 to the reboiler
 - d. Pump 3 (P-404): Bottoms product pump from C-401
2. Heat exchangers – System used to transfer heat between two or more fluids
 - a. Heat Exchanger 1 (E-401): Used to condense the waste distillate stream from C-401
 - b. Heat Exchanger 3 (E-402): Fan cooler to cool waste product from high temperatures
 - c. Heat Exchanger 4 (E-403): Heat exchanger to further cool the waste product from C-401
 - d. Heat Exchanger 5 (E-404): Reboiler for C-401
 - e. Heat Exchanger 5 (E-405): Fan cooler to cool biodiesel product stream from high temperatures
 - f. Heat Exchanger 6 (E-406): Heat exchanger to further cool the biodiesel product from C-401
3. Columns – A tall metal cylinder internally fitted with perforated horizontal plates or fillings to promote phase separation
 - a. Trayed Distillation Column (T-401): Separates unreacted oil from biodiesel products
4. Vessels – Store and dispense temperature sensitive materials
 - a. Reflux Drum (V-401): A horizontal vessel that temporarily stores distillate from T-301 before sending it back into the column
5. Tanks – Store and materials of a process
 - b. Tank 1 (TK-401): Storage tank for waste product
 - c. Tank 2 (TK-402): Storage tank for the biodiesel product
6. Filters – a device used to remove impurities or solid particles from liquids or gases
 - a. Filter (F-401): A prove up cartridge filter used to obtain the desired purity of the biodiesel product

Appendix IV (Updated Mass Balance)

Table 8. Detailed PFD Drawing mass balance on all streams taken from Aspen simulation.

Stream Number/ID	Units	S1	S1a	S2	S3	S3a	S4	S4a	S4b	S4c	S4d
From		TK-102	P-101	S-101	TK-102	P-102	S2/S3	P-103	P-104	P-105	P-106
To		P-101	S-101	S4	P-102	S4	P-103	P-104	P-105	P-106	F-101
Temperature	°F	77	179	179	77	77	139	149	160	170	180
Pressure	PSIA	14.7	40	34	15	40	34	725	1450	2176	2901
Vapor Fraction		0	0	0	0	0	0	0	0	0	0
Liquid Fraction		1	1	1	1	1	1	1	1	1	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Enthalpy	MMBTU/HR	-7.91	-7.909	-7.909	-3.16	-3.16	-44.72	-44.62	-44.52	-44.41	-44.31
Density	LB/CUFT	56.67	55.53	55.53	56.67	55.53	35.18	34.91	34.63	34.35	34.07
Total Mass Flow	LB/HR	8696	8696	8696	988	988	20385	20385	20385	20385	20385
Total Volumetric Flow	CUFT/HR	153	157	157	20	20	579	584	589	593	598
Component Mass Flow											
METHANOL	LB/HR	0	0	0	988	988	11688	11688	11688	11688	11688
WATER	LB/HR	0	0	0	0	0	0	0	0	0	0
GLYCEROL	LB/HR	0	0	0	0	0	0	0	0	0	0
BIODIESEL	LB/HR	0	0	0	0	0	0	0	0	0	0
USED COOKING OIL	LB/HR	8696	8696	8696	0	0	8696	8696	8696	8696	8696

Table 9. Detailed PFD Drawing mass balance on all streams taken from Aspen simulation.

Stream Number/ID	Units	S5	S6	S6a	S6b	S6c	S6d	S6e	S7	S7a	S8
From		F-101	R-101a/R-101b	VLV-201	VLV-202	VLV-203	E-201	E-204	V-201	P-201	V-201
To		R-101a/R-101b	VLV-201	VLV-202	VLV-203	E-201	E-204	V-201	P-201	C-201	E-202
Temperature	°F	1200	1200	1200	1200	1200	191	158	158	158	158
Pressure	PSIA	2890	2901	1939	977	36	26	15	15	29	15
Vapor Fraction		0.97	1	1	1	1	0.80	0.75	0	0	1
Liquid Fraction		0.03	0	0	0	0	0.20	0.25	1	1	0
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Enthalpy	MMBTU/HR	-32.88	-27.12	-27.12	-27.12	-27.12	-40.56	-41.14	-17.10	-17.10	-24.04
Density	LB/CUFT	42.51	32.61	36.14	39.70	43.14	51.39	52.05	52.05	52.05	
Total Mass Flow	LB/HR	20385	20385	20385	20385	20385	20385	20385	11377	11377	9008
Total Volumetric Flow	CUFT/HR	2451	2302	3443	6831	182976	81823	126973	219	219	194
Component Mass Flow											
METHANOL	LB/HR	11688	10753	10753	10753	10753	10753	10753	1745	1745	9008
WATER	LB/HR	0	0	0	0	0	0	0	0	0	0
GLYCEROL	LB/HR	0	896	896	896	896	896	896	896	896	0
BIODIESEL	LB/HR	0	8655	8655	8655	8655	8655	8655	8655	8655	0
USED COOKING OIL	LB/HR	8696	80	80	80	80	80	80	80	80	0

Table 10. Detailed PFD Drawing mass balance on all streams taken from Aspen simulation.

Stream Number/ID	Units	S8a	S9	S9a	S10	S10a	S11	S11a	S12	S12a	S13
From		E-202	V-202	P-202	C-201	E-203	V-203	P-203	V-203	P-204	S12a/S9a
To		V-202	P-202	S13	E-203	V-203	P-203	C-201	P-204	S13	P-107
Temperature	°F	148	148	148	148	148	148	148	148.2	148.3	148.3
Pressure	PSIG	15	15	29	15	15	15	29	14.7	29.01	29.01
Vapor Fraction		0	0	0	1	0	0	0	0	0	0
Liquid Fraction		1	1	1	0	1	1	1	1	1	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Enthalpy	MMBTU/HR	-28.33	-28.33	-28.33	-4.59	-5.40	-5.40	-5.40	-5.32	-5.32	-33.65
Density	LB/CUFT	46.45	46.45	46.44	N/A	46.45	46.45	46.44	46.45	46.44	46.44
Total Mass Flow	LB/HR	9008	9008	9008	1717	1717	1717	1717	1692	1692	10700
Total Volumetric Flow	CUFT/HR	194	194	194	23791	37	37	37	36	36	230
Component Mass Flow											
METHANOL	LB/HR	9008	9008	9008	1717	1717	1717	1717	1692	1692	10700
WATER	LB/HR	0	0	0	0	0	0	0	0	0	0
GLYCEROL	LB/HR	0	0	0	0	0	0	0	0	0	0
BIODIESEL	LB/HR	0	0	0	0	0	0	0	0	0	0
USED COOKING OIL	LB/HR	0	0	0	0	0	0	0	0	0	0

Table 11. Detailed PFD Drawing mass balance on all streams taken from Aspen simulation.

Stream Number/ID	Units	S13a	S14	S14a	S15	S16	S16a	S16b	S17	S17a	S18
From		P-107	C-201	P-206	E-205	E-206	P-206	E-301	V-301	P-302	V-301
To		S4	P-206	E-205	C-201	P-206	E-301	V-301	P-302	TK-301	P-301
Temperature	°F	148	159	159	303	299	299	78	78	78	78
Pressure	PSIA	40	15	40	40	15	29	29	15	29	14.7
Vapor Fraction		0	0	0	0.50	0	0	0	0	0	0
Liquid Fraction		1	1	1	0.50	1	1	1	1	1	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Enthalpy	MMBTU/HR	-33.65	-16.62	-16.62	-15.12	-11.08	-11.08	-12.23	-2.87	-2.87	-9.34
Density	LB/CUFT	46.44	52.13	52.12	48.66	48.73	48.73	54.08	78.65	78.64	53.89
Total Mass Flow	LB/HR	10700	11233	11233	11233	9684	9684	9684	922	922	8762
Total Volumetric Flow	CUFT/HR	230	215	216	9257	199	199	179	12	12	163
Component Mass Flow											
METHANOL	LB/HR	10700	1561	1561	1561	53	53	53	27	27	26
WATER	LB/HR	0	0	0	0	0	0	0	0	0	0
GLYCEROL	LB/HR	0	924	924	924	896	896	896	894	894	2
BIODIESEL	LB/HR	0	8668	8668	8668	8655	8655	8655	1	1	8654
USED COOKING OIL	LB/HR	0	80	80	80	80	80	80	0	0	80

Table 12. Detailed PFD Drawing mass balance on all streams taken from Aspen simulation.

Stream Number/ID	Units	S18a	S18b	S19	S19a	S20	S20a	S21	S21a	S22	S22a
From		P-301	E-302	Water	P-303	C-301	P-304	C-301	P-305	C-401	E-401
To		E-302	C-301	P-303	C-301	P-304	Waste	P-305	C-401	E-401	V-401
Temperature	°F	78	140	78	78	138	138	135	135	208	192
Pressure	PSIA	26	26	15	29	15	15	15	15	15	15
Vapor Fraction		0	0	0	0	0	0	0	0	1	0
Liquid Fraction		1	1	1	1	1	1	1	1	0	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Enthalpy	MMBTU/HR	-9.34	-9.06	-2.05	-2.05	-1.838	-1.84	-9.27	-9.27	-0.26	-0.31
Density	LB/CUFT	53.90	52.45	62.02	62.02	58.871	58.871	52.45	52.45	N/A	55.54
Total Mass Flow	LB/HR	8762	8762	300	300	283	283	8779	8779	50	50
Total Volumetric Flow	CUFT/HR	163	167	5	5	5	5	167	167	1272	1
Component Mass Flow											
METHANOL	LB/HR	26	26	0	0	19	19	7	7	7	7
WATER	LB/HR	0	0	300	300	262	262	38	38	43	43
GLYCEROL	LB/HR	2	2	0	0	2	2	0	0	0	0
BIODIESEL	LB/HR	8654	8654	0	0	0	0	8654	8654	0	0
USED COOKING OIL	LB/HR	80	80	0	0	0	0	80	80	0	0

Table 13. Detailed PFD Drawing mass balance on all streams taken from Aspen simulation.

Stream Number/ID	Units	S23	S23a	S24	S24a	S24b	S24c	S25	S25a	S26	S27
From		V-401	P-401	V-401	P-402	E-402	E-403	C-401	P-403	E-404	E-404
To		P-401	C-401	P-402	E-402	E-403	TK-401	P-403	E-404	C-401	P-404
Temperature	°F	192	192	192	192	192	78	168	168	397	397
Pressure	PSIA	15	29	14	29	23	18.51	15	40	40	15
Vapor Fraction		0	0	0	0	0	0	0	0	0.26	0
Liquid Fraction		1	1	1	1	1	1	1	1	0.74	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Enthalpy	MMBTU/HR	-3.10	-3.10	-0.24	-0.24	-0.24	-0.24	-10.19	-10.19	-8.89	-7.76
Density	LB/CUFT	55.54	55.53	55.54	55.53	55.53	57.10	51.31	51.31	45.91	51.31
Total Mass Flow	LB/HR	50	50	39	39	39	39	8992	8992	8992	8740
Total Volumetric Flow	CUFT/HR	1	1	1	1	1	1	175	175	2582	190
Component Mass Flow											
METHANOL	LB/HR	7	7	6	6	6	6	33	33	33	1
WATER	LB/HR	43	43	33	33	33	33	183	183	183	5
GLYCEROL	LB/HR	0	0	0	0	0	0	0	0	0	0
BIODIESEL	LB/HR	0	0	0	0	0	0	8695	8695	8695	8654
USED COOKING OIL	LB/HR	0	0	0	0	0	0	80	80	80	80

Table 14. Detailed PFD Drawing mass balance on all streams taken from Aspen simulation.

Stream Number/ID	Units	S27a	S27b	S27c	S28
From		P-404	E-405	E-406	S-401
To		E-405	E-406	S-401	TK-402
Temperature	°F	397	150	90	90
Pressure	PSIA	29	18	12	6
Vapor Fraction		0	0	0	0
Liquid Fraction		1	1	1	1
Solid Fraction		0	0	0	0
Enthalpy	MMBTU/HR	-7.76	-8.97	-9.24	-9.24
Density	LB/CUFT	51.31	51.31	51.31	51.31
Total Mass Flow	LB/HR	8740	8740	8740	8740
Total Volumetric Flow	CUFT/HR	190	167	163	163
Component Mass Flow					
METHANOL	LB/HR	1	1	1	1
WATER	LB/HR	5	5	5	5
GLYCEROL	LB/HR	0	0	0	0
BIODIESEL	LB/HR	8654	8654	8654	8654
USED COOKING OIL	LB/HR	80	80	80	80

Table 15. Detailed PFD Drawing mass balance on utility streams.

Stream Number/ID	Units	U1	U2	U3a & U3b	U4a & U4b	U5	U6	U7	U8	U9	U10
Component		Fuel Gas	Flue Gas	CW	CW	Air	Air	CW	CW	CW	CW
Temperature	°F	N/A	N/A	95	110	95	140	95	110	95	110
Pressure	PSIG	N/A	N/A	N/A	N/A	0	0	0	0	0	0
Vapor Fraction		1	1	0	0	1	1	0	0	0	0
Liquid Fraction		0	0	1	1	0	0	1	1	1	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Total Mass Flow	LB/MIN	13	13	3841	3841	16980	16980	640	640	4783	4783
Total Volumetric Flow	CFM	309	469	62	62	221958	2219562	10	11	78	79

Table 16. Detailed PFD Drawing mass balance on utility streams.

Stream Number/ID	Units	U11	U12	U13	U14	U15	U16	U17	U18	U19	U20
Component		CW	CW	MPS	MPS	Chilled W	Chilled W	LPS	LPS	CW	CW
Temperature	°F	95	110	414	301	42	67	318	106	95	110
Pressure	PSIG	0	0	275	275	0	0	73	73	0	0
Vapor Fraction		0	0	1	1	0	0	1	0	0	0
Liquid Fraction		1	1	0	0	1	1	0	1	1	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Total Mass Flow	LB/MIN	905	905	167	167	790	790	4	4	51	51
Total Volumetric Flow	CFM	15	15	3	3	3	13	22	0.1	0.83	0.84

Table 17. Detailed PFD Drawing mass balance on utility streams.

Stream Number/ID	Units	U21	U22	U23	U24	U25	U26	U27	U28	U29	U30
Component		HPS	HPS	Air	Air	CW	CW	Air	Air	CW	CW
Temperature	°F	482	187	85	140	85	110	85	140	85	110
Pressure	PSIG	566	566	0	0	0	0	0	0	0	0
Vapor Fraction		1	0	1	1	0	0	1	1	0	0
Liquid Fraction		0	1	0	0	1	1	0	0	1	1
Solid Fraction		0	0	0	0	0	0	0	0	0	0
Total Mass Flow	LB/MIN	56	56	2	2	93	93	1511	1511	179	179
Total Volumetric Flow	CFM	1	1	29	32	0	0	20836	22940	3	3

Appendix V (Equipment List)

Table 18. Equipment list and sizing for PFD Drawing 1.

Equipment Number	Name	PDF #	Description	Metalurgy	Motor Horsepower	Duty (MBtu/hr)* (Mbtu = 1000 Btu)	Operating Pressure (psig)	Operating Temperature (°F)	Design Pressure (psig)	Design Temperature (°F)
S-101	Feed Oil Filter	DWG-01	Cartridge filter	316 SS	N/A	N/A	Atmospheric	Atmospheric	N/A	N/A
TK-101	Used Cooking Oil Feed Storage Tank	DWG-01	Dome roof tank Volume: 200,000 gallons Height: 37.8 ft Diameter: 30 ft Thickness: 3.5"	CS	N/A	N/A	Atmospheric	Atmospheric	15" water gauge	700
TK-102	Methanol Feed Storage Tank	DWG-01	Dome roof tank Volume: 25100 gallons Height: 29.7 ft Diameter: 12 ft Thickness: 0.48"	CS	N/A	N/A	14.31	Atmospheric	15.8	700
P-101	Feed Oil Pump	DWG-01	Goulds HT 3196 Centrifugal Pump Rated for 19.5 gpm and 62.6 ft head	CS	0.375	N/A	25.19	179.21	450	700
P-102	Feed Methanol Pump	DWG-01	Goulds HT 3196 Centrifugal Pump Rated for 2.5 gpm and 72.4 ft head	CS	0.0483	N/A	25.19	77.19	450	700
P-103	Reactor Feed Pump	DWG-01	Goulds 7200CB (BB5) Barrel Multistage Pump Rated for 73 gpm and 1905.3 ft head	316 SS	39	N/A	710.49	149.17	4000	800
P-104	Reactor Feed Pump	DWG-01	Goulds 7200CB (BB5) Barrel Multistage Pump Rated for 73 gpm and 1986.5 ft head	316 SS	41	N/A	1435.68	159.71	4000	800
P-105	Reactor Feed Pump	DWG-01	Goulds 7200CB (BB5) Barrel Multistage Pump Rated for 74 gpm and 1970.3 ft head	316 SS	41	N/A	2160.87	170.06	4000	800
P-106	Reactor Feed Pump	DWG-01	Goulds 7200CB (BB5) Barrel Multistage Pump Rated for 74 gpm and 2002 ft head	316 SS	42	N/A	2886.06	180.21	4000	800
P-107	Methanol Recycle Pump	DWG-01	Goulds HT 3196 Centrifugal Pump Rated for 29 gpm and 31.3 ft head	CS	0.243	N/A	25.19	154	450	700
R-101a	Transesterification CSTR	DWG-01	Volume: 5731 gallons Diameter: 9.75 ft Height: 10.25 ft Thickness: 0.886 ft Jacketed with cooling water (spiral baffled jacket) Jacket Surface Area: 106.2 ft ²	316 SS	N/A	5761	2886.06	1200	3174.7	1700
R-101b	Transesterification CSTR	DWG-01	Volume: 5731 gallons Diameter: 9.75 ft Height: 10.25 ft Thickness: 0.886 ft Jacketed with cooling water (spiral baffled jacket) Jacket Surface Area: 106.2 ft ²	316 SS	N/A	5761	2886.06	1200	3174.7	1700
AG-101a	R-101a Agitator	DWG-01	Disc-mounted flat-blade paddle turbine impeller with a 7 ft shaft and a 6 ft impeller diameter	316 SS	59 hp	N/A	2886.06	1200	3174.7	1700
AG-101b	R-101b Agitator	DWG-01	Disc-mounted flat-blade paddle turbine impeller with a 7 ft shaft and a 6 ft impeller diameter	316 SS	59 hp	N/A	2886.06	1200	3174.7	1700
F-101	Reactor Feed Gas Preheater	DWG-01	Fired gas heater with radiant coils 3/4" tubes (0.065" thick)	316 SS	N/A	11433	2875.18	1200	3300 (pipes)	1700

Table 19. Equipment list and sizing for PFD Drawing 2.

Equipment Number	Name	PFD #	Description	Metalurgy	Motor Horsepower	Duty (MBtu/hr)* (Mbtu = 1000 Btu)	Operating Pressure (psig)	Operating Temperature (°F)	Design Pressure (psig)	Design Temperature (°F)
VLV-201	Product Throttle Valve	DWG-02	Not Major Equipment	316 SS	N/A	N/A	1924	1239.62	2116	1700
VLV-202	Product Throttle Valve	DWG-02	Not Major Equipment	316 SS	N/A	N/A	962	1222.9	1058	1700
VLV-203	Product Throttle	DWG-02	Not Major Equipment	316 SS	N/A	N/A	10.88	1200	12	1700
V-201	Methanol Flash Column	DWG-02	498 gallons Height: 11.07 ft Diameter: 2.78 ft Thickness: 0.2"	CS	N/A	N/A	Atmospheric	158	15" water gauge	700
V-202	Flash Column Reflux Drum	DWG-02	242 gallons Height: 5.48 ft Diameter: 2.76 ft Thickness: 0.2"	CS	N/A	N/A	Atmospheric	148.16	15" water gauge	700
V-203	Distillation Column Reflux Drum	DWG-02	46.1 gallons Height: 3.15 ft Diameter: 1.6 ft Thickness: 0.2"	CS	N/A	N/A	Atmospheric	148.16	15" water gauge	700
C-201	Methanol Distillation Column	DWG-02	Tray Number: 8 trays (assuming 64% efficiency) Tray type: Sieve Column Height: 30.4 ft Column Diameter: 1.14 ft Thickness: 0.2"	CS	N/A	N/A	Atmospheric	299.21	15.8	700
E-201	Reactor Effluent Fan Cooler	DWG-02	Air Fin Fan Cooler (45° pitch with G-fins) 1" Tubes with 20:1 fin to tube area ratio (4.36" fin diameter) Surface Area Tubes: 236.7 ft ² Surface Area Fins: 4734 ft ²	316 SS	Fan: 373 *Assuming 1" W.G. increase in pressure	13448	21.76	1201	24	1700
E-202	Flash Column Condenser	DWG-02	Shell and tube (U-tube) with cooling water Surface Area: 567.6 ft ²	Shell: CS Tube: CS	N/A	4294	Atmospheric	158	15.8	700
E-203	Distillation Column Condenser	DWG-02	Shell and tube (U-tube) with cooling water Surface Area: 120 ft ²	Shell: CS Tube: CS	N/A	813	Atmospheric	148.16	15.8	700
E-204	Flash Feed Cooler	DWG-02	Shell and tube (U-tube) with cooling water Surface Area: 57.75 ft ²	Shell: CS Tube: CS	N/A	575	10.88	191.09	12	700
E-205	C-201 Reboiler	DWG-02	Kettle reboiler Surface Area: 85.25 ft ²	Shell: CS Tube: CS	N/A	1491	275.38	413.6	303	700
P-201	Flash Column Bottoms Pump	DWG-02	Goulds LF 3196 Centrifugal Pump Rated for 27.25 gpm and 37.5 ft head	316 SS	0.303	N/A	14.31	158.12	450	700
P-202	Methanol Distillate Pump 1	DWG-02	Goulds LF 3196 Centrifugal Pump Rated for 24.2 gpm and 41.6 ft head	316 SS	0.27	N/A	14.31	148.25	450	700
P-203	Reflux Pump	DWG-02	Goulds LF 3196 Centrifugal Pump Rated for 4.61 gpm and 41.3 ft head	316 SS	0.051	N/A	14.31	148.25	450	700
P-204	Methanol Distillate Pump 2	DWG-02	Goulds LF 3196 Centrifugal Pump Rated for 4.5 gpm and 42.3 ft head	316 SS	0.051	N/A	14.31	148.25	450	700
P-205	Distillation Column Bottoms Pump	DWG-02	Goulds LF 3196 Centrifugal Pump Rated for 26.9 gpm and 66.3 ft head	316 SS	0.53	N/A	25.19	159.29	450	700
P-206	Distillation Column Bottoms Product Pump	DWG-02	Goulds LF 3196 Centrifugal Pump Rated for 24.8 gpm and 36.6 ft head	316 SS	0.27	N/A	14	299.21	450	700

Table 20. Equipment list and sizing for PFD Drawing 3

Equipment Number	Name	PFD #	Description	Metalurgy	Motor Horsepower	Duty (MBtu/hr)* (Mbtu = 1000 Btu)	Operating Pressure (psig)	Operating Temperature (°F)	Design Pressure (psig)	Design Temperature (°F)
V-301	Glycerol Separation Decanter	DWG-03	223.3 gallons Height: 5.35 ft Diameter: 2.69 ft Thickness: 0.2"	CS	N/A	21	Atmospheric	78	15.8	700
C-301	Biodiesel Washing Column	DWG-03	Volume: 208.3 gallon Column Height: 8.3 ft Column Diameter: 2.09 ft Thickness: 0.2"	CS	N/A	N/A	14.7	140	15.7	700
TK-301	Glycerol Storage Tank	DWG-03	Volume: 14734 gallons Diameter: 9.75 ft Height: 10.25 ft Thickness: 0.886 ft	CS	N/A	N/A	Atmospheric	Atmospheric	15" water gauge	700
E-301	Decanter Feed Cooler	DWG-03	Shell and tube (U-tube) with chilled water Surface Area: 319.09 ft ²	Shell: CS Tube: CS	N/A	1155	Atmospheric	299.21	14.7	700
E-302	Washing Column Feed Heater	DWG-03	Shell and tube (U-tube) with low pressure steam Surface Area: 11.3 ft ²	Shell: CS Tube: CS	N/A	277	72.33	318.2	79.6	300
P-301	Decanter Top Product Pump	DWG-03	Goulds LF 3196 Centrifugal Pump Rated for 20.3 gpm and 28.8 ft head	316 SS	0.172	N/A	10.9	78	450	700
P-302	Glycerol Product Pump	DWG-03	Goulds LF 3196 Centrifugal Pump Rated for 1.5 gpm and 25.5 ft head	316 SS	0.016	N/A	14.31	78.09	450	700
P-303	Washing Water Feed Pump	DWG-03	Goulds LF 3196 Centrifugal Pump Rated for 0.6 gpm and 34.2 ft head	316 SS	0.0069	N/A	14.7	78.06	450	700
P-304	Washing Column Bottoms Pump	DWG-03	Goulds LF 3196 Centrifugal Pump Rated for 0.6 gpm and 0.9 ft head	316 SS	0.00018	N/A	Atmospheric	138.36	450	700
P-305	Washing Column Top Pump	DWG-03	Goulds LF 3196 Centrifugal Pump Rated for 20.9 gpm and 0.5 ft head	316 SS	0.00312	N/A	Atmospheric	134.87	450	700

Table 21. Equipment list and sizing for PFD Drawing 4

Equipment Number	Name	PFD #	Description	Metalurgy	Motor Horsepower	Duty (MBtu/hr)* (Mbtu = 1000 Btu)	Operating Pressure (psig)	Operating Temperature (°F)	Design Pressure (psig)	Design Temperature (°F)
V-401	C-401 Reflux Drum	DWG-04	Volume: 1.23 gallons Height: 0.94 ft Diameter: 0.49 ft Thickness: 0.2"	CS	N/A	N/A	Atmospheric	193.2	15.8	700
C-401	Biodiesel Distillation Column	DWG-04	Tray Number: 10 trays (assuming 64% efficiency) Tray type: Sieve Column Height: 35 ft Column Diameter: 0.68 ft Thickness: 0.20"	CS	N/A	N/A	Atmospheric	397.14	15.7	700
TK-401	Waste Storage Tank	DWG-04	Volume: 807 gallons Diameter: 5 ft Height: 5.5 ft Thickness: 0.2"	CS	N/A	N/A	14.7	Atmospheric	15.8	700
TK-402	Biodiesel Stage Tank	DWG-04	Volume: 203885 gallons Diameter: 30 ft Height: 33.9 ft Thickness: 0.6"	CS	N/A	N/A	Atmospheric	Atmospheric	15" water gauge	700
S-401	Biodiesel Product Cartridge Filter	DWG-04	High Flow Cartridge Filter	316 SS	N/A	N/A	Atmospheric	Atmospheric	N/A	N/A
E-401	C-401 Condesner	DWG-04	Shell and tube (U-tube) with cooling water Surface Area: 3.14 ft ²	Shell: CS Tube: CS	N/A	46	Atmospheric	208.2	14.7	700
E-402	C-401 Waste Product Stream Fan Cooler	DWG-04	Air Fin Fan Cooler (45° pitch with G-fins) 1" Tubes with 20:1 fin to tube area ratio Surface Area Tubes: 0.195 ft ² Surface Area Fins: 3.9 ft ²	CS	Fan: 0.0031 *Assuming 1" W.G. increase in pressure	1.7	14.7	192.9	15.8	700
E-403	Biodiesel Product Cooler	DWG-04	Shell and tube (U-tube) with cooling water Surface Area: 0.918 ft ²	CS	N/A	2.3	14.7	150	15.8	700
E-404	C-401 Reboiler	DWG-04	Kettle Reboiler 3/4" tubes (0.049" thick) Surface Area: 190.85 ft ²	316 SS	N/A	1304	564.46	482	2400 (pipes)	1000
E-405	Biodiesel Product Air Fan Cooler	DWG-04	Air Fin Fan Cooler (45° pitch with G-fins) 1" Tubes with 20:1 fin to tube area ratio Surface Area Tubes: 57.64 ft ² Surface Area Fins: 1152.8 ft ²	CS	Fan: 2.2 *Assuming 1" W.G. increase in pressure	1206	14.3	397.26	15.7	700
E-406	Biodiesel Product Cooler	DWG-04	Shell and tube (U-tube) with cooling water Surface Area: 105.86 ft ²	CS	N/A	267	14.3	150	15.7	700
P-401	C-401 Reflux Pump	DWG-04	Goulds LF 3196 Centrifugal Pump Rated for .113 gpm and 35.2 ft head	316 SS	1.29E-03	N/A	14.7	192.2	450	700
P-402	C-401 Top Product Pump	DWG-04	Goulds LF 3196 Centrifugal Pump Rated for 0.09 gpm and 33.9 ft head	316 SS	9.90E-04	N/A	14.7	192.29	450	700
P-403	C-401 Bottom Pump	DWG-04	Goulds LF 3196 Centrifugal Pump Rated for 21.85 gpm and 66.8 ft head	316 SS	0.43	N/A	25.19	158.49	450	700
P-404	C-401 Bottom Product Pump	DWG-04	Goulds LF 3196 Centrifugal Pump Rated for 23.7 gpm and 37.8 ft head	316 SS	0.264	N/A	14.31	397.13	450	700

Appendix VI (Hand Calculations and Discussion)

The Fenske equation is used as a short-cut method to hand-calculate the minimum number of trays needed in a fractional distillation column at total reflux, including the reboiler. The equation is as follows:

$$N = \frac{\log \left[\left(\frac{LK_d}{HK_d} \right) \left(\frac{HK_b}{LK_b} \right) \right]}{\log \alpha_{avg}} \quad (1)$$

Where,

N = the minimum number of theoretical trays (including the reboiler) required at total reflux.

LK = light key component mole fraction

HK = heavy key component mole fraction

d = distillate

b = bottoms

α = relative volatility

The relative volatility of the light key to the heavy is not constant from the top of the column to the bottom of the column. [63] Therefore, the average relative volatility is calculated with the following equation:

$$\alpha_{avg} = \sqrt{\alpha_{LKd} \alpha_{LKb}} \quad (2)$$

Relative volatility can be calculated from Henry's Law and using a ratio of K-values to eliminate the temperature dependence on the relative volatility. [64]

$$K = \frac{y_i}{x_i} \quad (3)$$

$$\alpha = \frac{K_{LK}}{K_{HK}} \quad (4)$$

Where,

y_i = vapor mole fraction of component i

x_i = liquid mol fraction of component i

The biodiesel distillation column, C-401, is designed to purify the biodiesel product from any impurities including any methanol, water, and excess oil. See Figures 26 and 27. It will be used for hand-calculation of theoretical number of stages, theoretical number of trays, actual number of trays using Murphree Efficiency of 64%, column height, diameter, and thickness. These dimensions will then be compared to the values obtained from the Aspen Plus simulation.

The light key is the methanol and the heavy key is defined as the biodiesel product. From the Aspen simulation LK_b , HK_b , LK_d , HK_d , and K-values for the LK and HK in the distillate and bottoms are obtained. See Tables 22, 23, and 24.

Table 22. Flow rates and mol fractions of light key and heavy key in distillate and bottoms.

Distillate			Bottoms		
Molar flow rate (D)	2.00617066	lbmol/hr	Molar flow rate (B)	29.58707	lbmol/hr
LK	0.08679717		LK	1.36E-03	
HK	2.90E-06		HK	0.986499	

Table 23. K-values for light key and heavy key in distillate.

distillate (waste)	
component	K value
methanol	4.45488227
biodiesel	135.335466

Table 24. K-values for light key and heavy key in bottoms.

bottoms (product)	
component	K value
methanol	65.9685711
biodiesel	0.01276106

$$\alpha_{LKd} = \frac{4.45488227}{135.335466} = 0.032$$

$$\alpha_{LKb} = \frac{65.9685711}{0.01276106} = 5169.52$$

$$\alpha_{avg} = \sqrt{(0.032)(5169.52)} = 13.04$$

$$N = \frac{\log\left[\left(\frac{0.0868}{0.0000026} \cdot 2.0062 \frac{\text{lbmol}}{\text{hr}}\right) \left(\frac{0.9865}{0.00136} \cdot 29.587 \frac{\text{lbmol}}{\text{hr}}\right)\right]}{13.04} = 6.58 \text{ theoretical stages or } 5.58 \text{ theoretical trays}$$

Rounding up, the Fenske equation gives 7 theoretical stages and 6 theoretical trays if the reboiler is not included. If 6 theoretical trays are divided by the Murphree Efficiency of 64% for the column, 9 actual trays are needed for this column.

$$\frac{5.58}{0.64} = 8.7 = 9 \text{ actual trays}$$

If the tray spacing is estimated at 2.3 feet, then the height of the column can be calculated.

$$\text{Height} = 9 \text{ trays} \cdot 2.3 \text{ feet} = 20.7 \text{ feet or } 21 \text{ feet}$$

Six extra feet must be added to the top and bottom of the column to account for the dome and sump. Therefore, the calculated height of the column is 32.7 feet or 33 feet.

To calculate the diameter of the column, the average residence time of the column is obtained from Aspen. See Table 25.

Table 25. Downcomer residence time from Aspen simulation

Downcomer Residence Time (hr)	
	0.327888228
	0.00039429
	0.000394259
	0.000393476
Average	0.082267563

$$Diameter = \left(\frac{2 * volumetric\ flow\ rate * residence\ time}{\pi} \right)^{\frac{1}{3}} = \left(\frac{2 * 167.378 \frac{ft^3}{hr} * 0.0822\ hr}{\pi} \right)^{\frac{1}{3}} = 2.06\ feet$$

The thickness of the column is designed to accommodate the pressure that the column operates at. A safety of 10% is added onto the optimal operating pressure for thickness design purposes.

$$thickness = \frac{1.10 * pressure * diameter}{2 * stress\ factor\ of\ steel - 1.2 * 1.10 * pressure}$$

$$thickness = \frac{1.10 * 202650.096\ Pa * 0.628\ meters}{2 * 8894550\ Pa - 1.2 * 1.10 * 202650.096\ Pa} = 0.000789\ meter\ or\ 0.031052\ inches$$

Table 26 compares these values to the values obtained directly from the Aspen Plus simulation.

Table 26. Aspen column dimensions vs hand-calculated dimensions.

	Units	Aspen Values	Fenske
Stages		8	7
Trays		10	9
Height	feet	35	33
Diameter	feet	0.68	2.06
Thickness	inches	0.0099	0.00079

(It is noted that for the thickness obtained for the column in both the hand-calculation and Aspen, it is less than the standard minimum of 7 mm thickness for any pressure column. For the purposes of proper designing of this column, the thickness is set at 0.28 inches.) Aspen and the Fenske equation are very close in the determination of the number of stages, trays, and height required for the distillation column. The main differences are noted in the values for the

diameter and thickness of the column. Aspen determined the diameter to be about 1.4 feet less than the hand-calculation, and the thickness about 0.00911 inches more than the hand calculation.

The differences between the Fenske method and the Aspen simulation can be analyzed. The values Aspen calculates are only as good as the information supplied to Aspen, including the equation of state and property methods used. In this case, UNIQUAC base method is used for C-401 distillation column. Aspen also goes through a tray-by-tray rigorous calculation to obtain actual equilibrium data for each component based on that tray's operating temperature and pressure. This data takes into account different interactions between each component on a molecular level including Van der Waals interactions and differences in polarity.

The short-cut methods such as the Fenske equation can be used to see if the rigorous Aspen calculation makes sense. In this case, the number of stages, the number of actual trays, and the height of the column are very close to each other. The thickness for both the Fenske and Aspen results is a relatively small thickness which will be defaulted to a minimum thickness of 7 mm. The main difference is in the diameter of the column. A difference of 1.4 feet is noted between the two values. This might make a difference in the cost and efficiency of the column and should be investigated further.

Appendix VII (Preliminary Utilities)

Table 27. Summary of total utilities usage.

Description	Type	Rate	Units
Cooling Air	Air	423,467	CFM
Chilled Water	Water	154	GPM
Cooling Water	Water	2,385	GPM
Wash Water	Water	0.60	GPM
Electricity	Power	818,660	Wh
Natural Gas	Gas	309	CFM
Low Pressure Steam (LPS)	Steam	515	lb/hr
Medium Pressure Steam (MPS)	Steam	3,060	lb/hr
High Pressure Steam (HPS)	Steam	2,951	lb/hr

Table 28. Summary of utility specifications. [65]

Utility	Specifications
Cooling Air	85 °F
Chilled Water	42°F
Cooling Water	85 °F
Wash Water	85°F
Natural Gas	heat content: 1,029 Btu per cubic foot [66]
LPS	73 psig (6 bar), 318°F
MPS	275 psig (20 bar), 414°F
HPS	565 psig (40 bar), 482°F

Table 29. Utilities Summary for PFD Drawing 1, reactor section.

Equipment	Utility	Usage	Operating Conditions Description
P-101	Electricity	466 Wh	Pump 8,696 lb/hr of used cooking oil from stream S1 to unit S-101
P-102	Electricity	60 Wh	Pump 988 lb/hr of methanol from stream S3 to S4
P-103	Electricity	48,471 Wh	Pump 20,385 lb/hr feed from stream S4 to S4a
P-104	Electricity	50,956 Wh	Pump 20,385 lb/hr feed from stream S4a to S4b
P-105	Electricity	50,956 Wh	Pump 20,385 lb/hr feed from stream S4b to S4c
P-106	Electricity	52,199 Wh	Pump 20,385 lb/hr feed from stream S4c to unit F-101
P-107	Electricity	302 Wh	Pump 10,700 lb/hr methanol recycle from stream S13 to S4
R-101a	Cooling Water	792 GPM	Maintain temperature of R-101a at 1200°F
R-101b	Cooling Water	792 GPM	Maintain temperature of R-101b at 1200°F
AG-101a	Electricity	73,327 Wh	Agitate 2,451 ft ³ /hr in R-101a
AG-101b	Electricity	73,327 Wh	Agitate 2,451 ft ³ /hr in R-101b
F-101	Natural Gas	309 CFM	Heat 20,385 lb/hr feed in stream S4d from 180°F to 1200°F

Table 30. Utilities Summary for PFD Drawing 2, methanol recycle section.

Equipment	Utility	Usage	Operating Conditions Description
E-201	Electricity Cooling Air	463,577 Wh 2,786,367 CFM	Cool 20,385 lb/hr in stream S6c from 1200°F to 190°F
E-202	Cooling Water	574 GPM	Cool 9,008 lb/hr in stream S8 from 158°F to 148°F
E-203	Cooling Water	109 GPM	Condense 1,717 lb/hr in stream S10 from vapor to liquid
E-204	Cooling Water	77 GPM	Cool 20,385 lb/hr in stream S6d from 191°F to 158°F
E-205	MPS	3060 lb/hr	Heat 11,233 lb/hr in stream S14a from 159°F to 303°F
P-201	Electricity	377 Wh	Pump 11,377 lb/hr in stream S7 to unit C-201
P-202	Electricity	336 Wh	Pump 9,008 lb/hr methanol in stream S9 to methanol recycle stream S13
P-203	Electricity	64 Wh	Pump 1,717 lb/hr methanol reflux from stream S11 to unit C-201
P-204	Electricity	64 Wh	Pump 1,692 lb/hr methanol in stream S12 to methanol recycle stream S13
P-205	Electricity	659 Wh	Pump 11,233 lb/hr in stream S14 to unit E-205
P-206	Electricity	336 Wh	Pump 9,684 lb/hr in stream S16 to unit E-301

Table 31. Utilities Summary for PFD Drawing 3, biodiesel washing section.

Equipment	Utility	Usage	Operating Conditions Description
C-301	Water	0.60 GPM	Wash 8,762 lb/hr from stream S18b
E-301	Chilled Water	154 GPM	Cool 9,684 lb/hr in stream S16a from 299°F to 78°F
E-302	LPS	515 lb/hr	Heat 8,762 lb/hr in stream S18a from 78°F to 140°F
P-301	Electricity	214 Wh	Pump 8,762 lb/hr in stream S18 to unit E-302
P-302	Electricity	20 Wh	Pump 922 lb/hr in stream S17 to Glycerol storage tank
P-303	Electricity	9 Wh	Pump 300 lb/hr water in stream S19 to unit C-301
P-304	Electricity	0.20 Wh	Pump 283 lb/hr in stream S20 to waste storage tank
P-305	Electricity	4 Wh	Pump 8,779 lb/hr in stream S21 to unit C-401

Table 32. Utilities Summary for PFD Drawing 4, biodiesel distillation section.

Equipment	Utility	Usage	Operating Conditions Description
E-401	Cooling Water	6 GPM	Cool 50 lb/hr in stream S22 from 208°F to 192°F
E-402	Electricity Cooling Air	4 Wh 49 CFM	Cool 39 lb/hr in stream 24a from 192°F to 150°F
E-403	Cooling Water	0.31 GPM	Cool 39 lb/hr in stream S24b from 150°F to 90°F
E-404	HPS	2951 lb/hr	Heat 8,922 lb/hr in stream S25a from 168°F to 397°F
E-405	Electricity Cooling Air	2734 Wh 34,847 CFM	Cool 8,740 lb/hr in stream S27a from 397°F to 150°F
E-406	Cooling Water	36 GPM	Cool 8,740 lb/hr in stream S27b from 150°F to 90°F
P-401	Electricity	1.6 Wh	Pump 50 lb/hr reflux in stream S23 to unit C-401
P-402	Electricity	1.23 Wh	Pump 39 lb/hr in stream S24 to unit E-402
P-403	Electricity	534 Wh	Pump 8,992 lb/hr in stream S25 to unit E-404
P-404	Electricity	328 Wh	Pump 8,740 lb/hr in stream S27 to unit E-405

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